→ AI is a branch of science which deals with helping machine finds solutions to complex problems in a more human-like fashion.

→ It involves drawing characteristics from Human and applying them as algorithms in a computer friendly way.

→ AI is not a system but it is implemented in the system.

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→ limitations of Human mind

→ AI is a popular branch of computer science that concerns with building intelligent machines to perform intelligent tasks.

- AI is a field of study that involves creating machines that can learn, reason and act like humans.

- They perform tasks with AI that involves datasets in large size or to analyse.

Artificial Intelligence is composed of 2 words Artificial defines man made i.e made or produced by human being rather than →

what AI will do.

- make predictions
- learn from experience
- Automate tasks
- perform complex tasks

AI will work based on concepts

① Machine learning

AI systems use ML to learn from data sets and improve their performance.

② Neural networks

AI systems use neural networks to learn complex relationships between inputs and outputs

③ Deep learning AI systems use deep learning to learn from data sets and improve their performance.

AI is a set of technologies that enable computers to perform a variety of advanced functions, including the ability to see, understand and translate spoken and written language, analyze data, make recommendations and more.

4. NLP
5. Robotics
6. Cognitive Computing
7. Expert systems

(capabilities human like reasoning to render give

AI means a man made thinking power.

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Name :
Year & Sem (2) Subject : AI
H.T. No. :
Date :
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AI works by simulating human intelligence through the use of algorithms data and computational power.

- Goal is to enable machines or s/w to perform tasks that typically require human intelligence, such as learning, reasoning, problem-solving, perception and language understanding.

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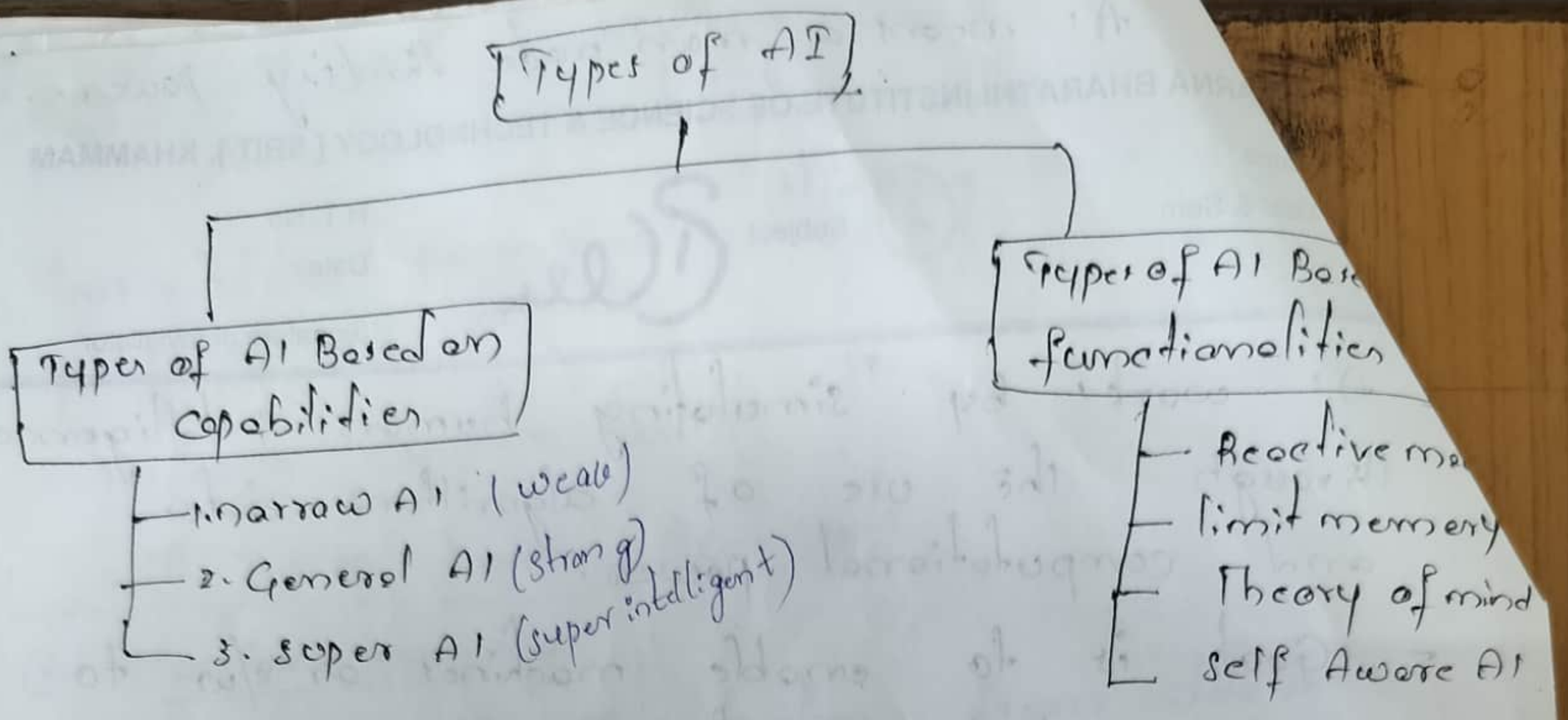
Father of AI

Greatest innovators in the field of AI was John McCarthy who got the title Father of Artificial Intelligence for his contribution to the field of computer science and AI.



There are different types of AI with each with its own strengths and weaknesses.

AI is divided into 3 primary types based on capabilities and four type based on functionalities.



1. narrow AI — voice assistants like Siri or Alexa that understand specific commands.
   - facial recognition — s/w used in security systems
   - Recommendation engines used by platforms like netflix or Amazon, chatbot, spam filters, image/speech recognition, autonomous cars
2. wide range of tasks can be considered.
   systems have capability to understand and learn
   - AI remains as a theoretical concept.
     potential applications.
     - Robots that can learn new skills and adapt to unforeseen challenges in real time.
     - If autonomously diagonse and solve complex medical issues across various specializations.

per Intelligence: Surpasses intelligence of
uman in solving problem, creativity and
over all abilities.
- super AI develops emotions, desires, need
and belief of their own.
- They are able to make decisions of their
own and solve problems of its own.
- These AI not only able to complete
tasks better then human but also under
and interpret emotions and respond in a
human like manner.

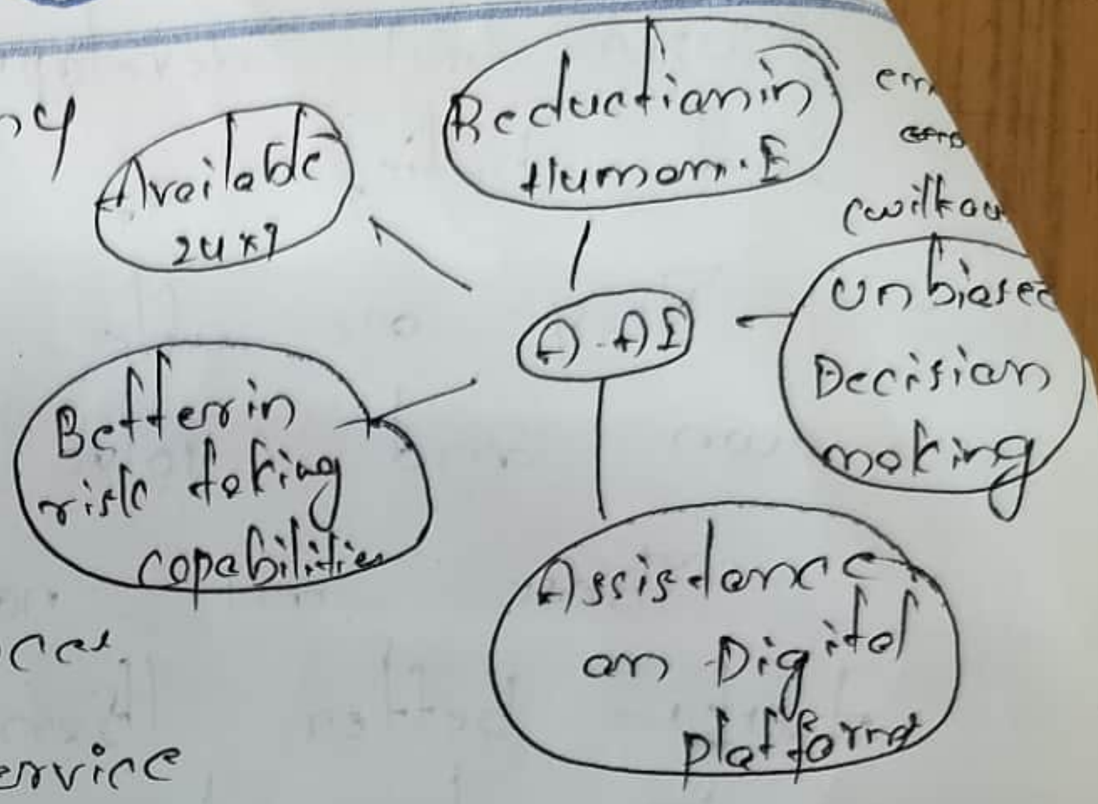
Functionalities:

1. Reactive machines are the most basic
form of AI.
They operate purely based on present data,
they do not store any previous experience
or learn from past actions.
- Here system respond to specific I/P's
with fixed O/P's and are unable to adapt.

2. limited memory in AI it will learn from
past data to improve future responses.
- most modern AI will come under this
- These system use historical data to
make decisions and predictions

Advantages of AI many

- Automation
- efficiency
- cost.
- personalized experiences
- Improved customer service
- Improved transportation
- Improved financial services
- " Agriculture
- " cyber security
- " scientific discovery
- " accessibility



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- High accuracy with less errors
- High speed
- High reliability
- useful for risky areas
- Digital Assistant-
- useful as a public utility.

Disadvantages

1. creativity
2. Emotional intelligence
3. encouraging Human laziness
4. privacy concerns.
5. Job Displacement

but do not have long term memory present
ex: self driving cars - they use present
traffic movements and traffic signs.

chatbot that can remember recent
conversations to improve the flow
and relevance of replies.

3. Theory of mind It aims to understand
human emotions, beliefs
intensions and desires.

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This type of AI remains in development
it would allow machines to engage in
more sophisticated interactions by
perceiving emotions and adjusting
behavior accordingly.

. Potential Applications

. Human robot interaction
where AI could detect emotions
and adjust its responses to empathize
with humans.

collaborative robots that work alongside human in fields like health care, adapting their tasks based on the needs of the patients.

4. Self-Awarenes AI

It is advanced stage of AI that possenes self consciousnes and awarenes.

It have the ability to not only understand and react to emotions but also have its own consciousnes sim to human awarenes.

potential Appli

. Fully autonomous systems that can make moral and ethical decisions.

. AI systems that can independently pursue goals based on their understanding of the world around them.

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Building AI systems
- perception < per acquires medical symptoms so on.. / vehicle signals and so on..
- Reasoning < knowledge reason Decision making
- Action < Inference knowledge representation, problem classification solving, game theory, what is sensed, N.L etc.
what model to be used

Biological systems interact with their actuation, speech etc.. means making decisions or taking actions based on

Rationally Intelligent system:

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logical rules, probability theory, optimization techniques.

system think like humans) rationally

" " rationally

system acts like humans rationally

" " rationally

- Think < Human like (cognitive science)
Rationally - laws of thought
- Act < Human - Turning test
Rationally - Rational Agent.

Agent will depend on (G perceiving (think) and acting.

Human Agent - eyes, ears ...
Artificial Agent - camera, infrared range finders
S/w Agent - keys strokes, file contents ...

Rationality:

Rationality in AI is the idea of making AI Agents as rational as possible.

- It is a central concept in AI.
- It is used to make autonomous programs that can make decisions based on logic and reasoning.

Rationality will work depending on these clarifications

① Decision making

RA use logic and reasoning to make decisions that optimize their behavior and achieve their goals.

② Adoptability : RA learn from their environment and adapt their behavior, which improves performance

3. Autonomy RA operate independently of human intervention, which can increase efficiency and reduce human error.

Components of Rationality

1. perception
2. ~~Automicity~~ Reasoning (to make decisions)
3. Action (Act based on decision)
4. Learning (RA learn from Action their and adapt behaviour.

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Intelligent Agent: An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through effectors

- A human agent has eyes, ears and other organs as for sensors
- hands, legs, mouth and other body parts for effectors/actuators.
- A robatic agent substitutes cameras, infrared signals.
- Intelligent agents are autonomous agents entities in AI that use sensors and actuators to achieve goals in their environment.
- They can learn from their environment and adapt to new information.

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characteristics: AI system are designed to simulate human int. & perform tasks

1. Autonomy: A will function dep according to human intervention
other s/w.
2. Reactivity: A detect changes in their environment and react accordingly.
3. proactivity: A can take initiative and perform goal related actions.
4. Social: A can work with human and other agents through speech recognition natural language and text meneging.

Intelligence means the ability of thinking
the ability of creating,
the ability of understanding.
the ability of classifying
the ability of adopting
the ability of solving problems
the ability of reasoning
the ability of learn from experience.
the ability to comprehend complex ideas
ability to solve problems
ability to use natural language fluently
ability to ~~pursue~~ perceive relationships
" to adapt new situations.

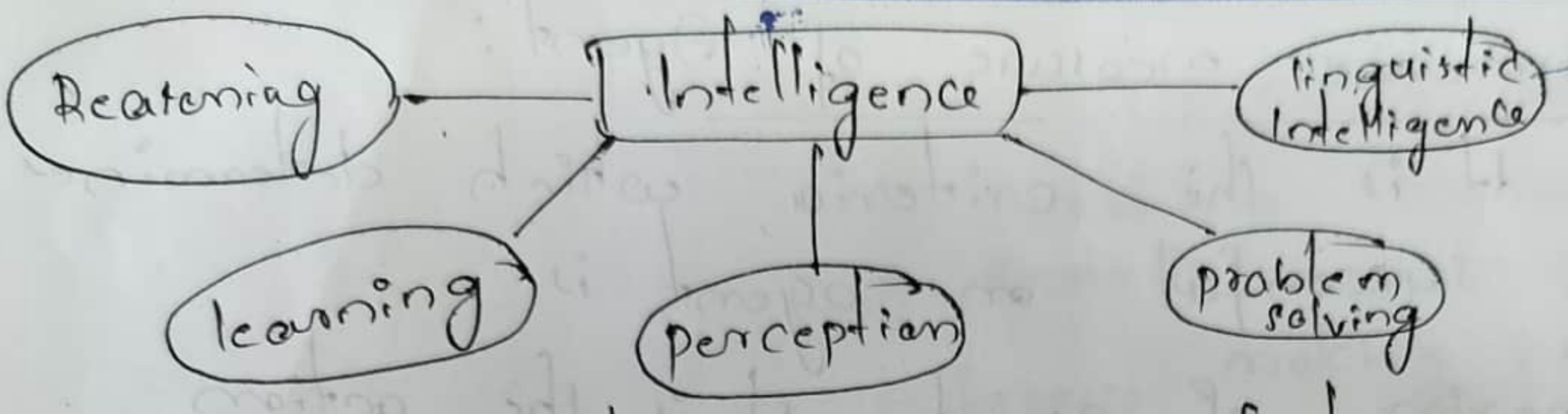
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(2) (10)



words can have many meanings and can form and be used in a variety of ways.

- An agent is anything that can be viewed as perceiving the environment through sensors and acting upon that environment through actuators.

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Human agent

- perceiving the environment can be done by senses.
- eyes, ears and other organs for sensors
- hands, legs, mouth and other body parts for actuators.

Robotic agent

- cameras & infrared range finders for sensors.
- various motors for actuators.

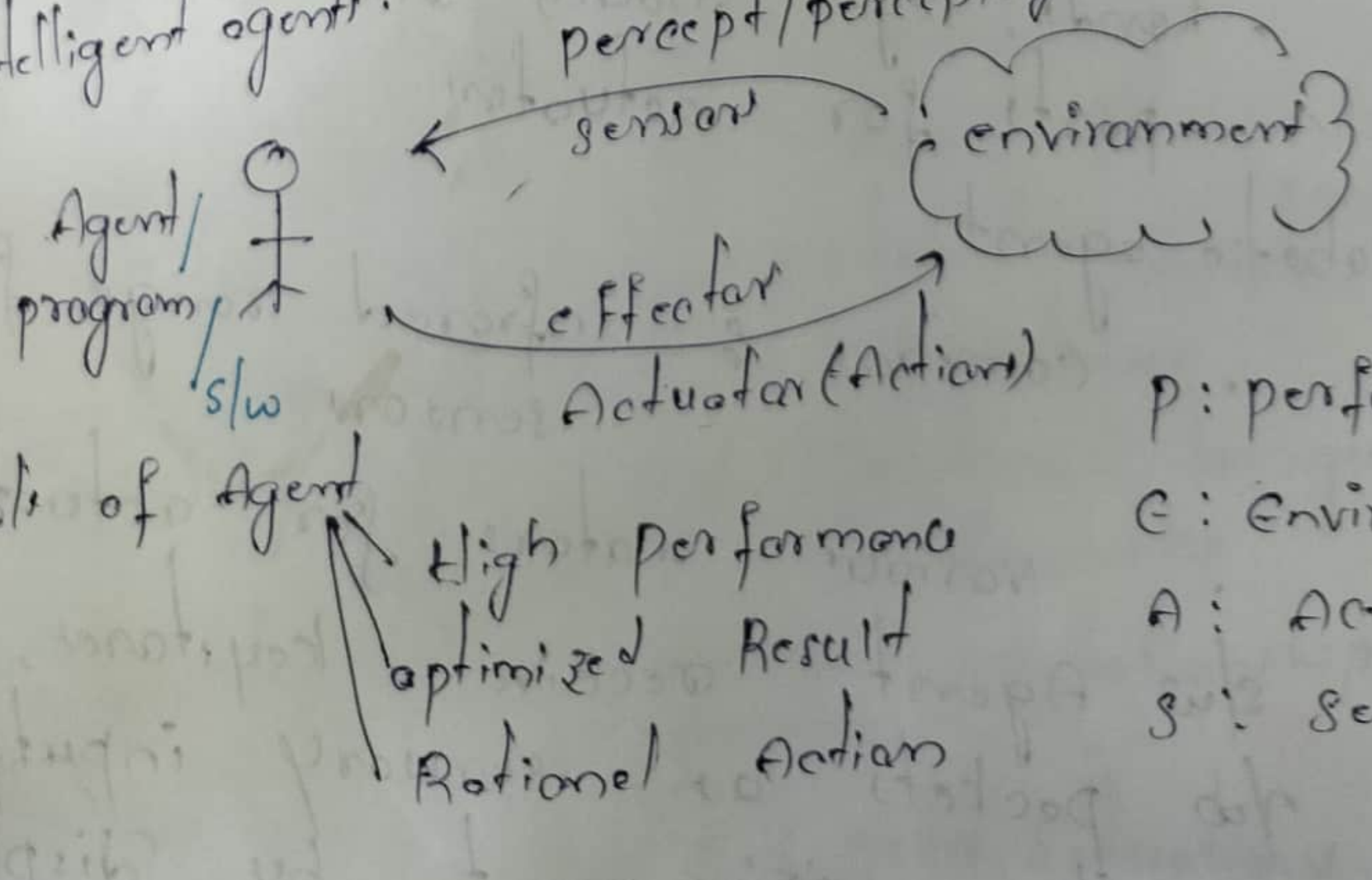
A s/w Agent receives keystrokes, file contents and s/w packets as sensory inputs and acts on the environment by displaying

on the screen, writing files and sending n/w packets.

Agent terminology:

1. performence measure of Agent:
   It is the criteria, which determines how succesful an agent is
2. Behavior of Agent It is the action that agent perform after any given sequence of percepts.
3. percept. It is agent's perceptual inputs at a given instance.
4. percept sequence. It is the history of all that an agent has perceived till date.
5. Agent function - It is a map from the precept sequence to an action.

Agents/ Intelligent agents.



Goals of Agent

- High performance
- optimized Result
- Rational Action

P: performance
E: Environment
A: Actions
S: Sensors

A R.A : in AI is an Autonomous system designed to ~~perform~~ perceive its environment, procen information and take actions to achieve the best possible outcomes or Expected utility, maximizing a performance measure based on its goals and available knowledge, much like a smart decision - maker.

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Rational Agent is a specific, superior type of I.A that always chooses the best action to maximize Expected performance, using logical reasoning to achieve optimal outcomes (like a chess AI or financial trading bot)

- All R.A are intelligent, but all I.A are perfectly rational
- An I.A can be designed to achieve a specific goal. But A R.A has a specific goal and optimizes its behaviour.

characteristics of I.A

Autonomy : Agent operates with out human intervention

Reactivity : It perceives and responds to change in environment.

proactiveness : It takes initiative toward achieving goals.

Social Ability : Some I.A interact with other Agents or humans.

Rationality means the ability of agent to make right decision.

Agent function & Agent program

Agent depends on function — depends on set of actions.

$f: P^* \rightarrow A$ possible

P^* is the set of all percept sequences.

A - ~~Action~~ set of actions.

~~Assuming~~ problem solving Agents:

Goals helps to organize the behaviour by limiting the objectives, the agent is trying to achieve.

- Goal formulation based on the current situation and agent performance measure. it is the first step in problem solving.

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- goal to be considered as the set of states in the world.
- problem formulation - It is the process of deciding what actions, and states to consider, given a goal.
- Search solution Execution:

This process of looking for such a sequence is called search

- the search algorithm takes problem as an input and returns a solution in the form of an action sequence.

ip.mic.gov.in//

- once a solution is found it be carried out, this is called the execution phase.

simple formulate, search, execute design for an agent.

→ After formulating a goal & problem to solve the agent calls a search procedure to solve it. It uses the solution to guide its actions.

well defined problems and solutions:

A problem can be defined formally by 4 components.

- initial state - where the agent starts first. In(state)
- successor function - possible no. of actions available to the agent.
- The most common formulation used a successor function.

x - is a state

SUCCESSOR - FN (x)

- returns a set of <action, successor> ordered pairs,
- where each action is one of the legal actions in state x and each successor is a state that can be reached from 'x' by applying action.
- The initial state and successor function implicitly define the state space of the problem.
- The set of all states reachable from initial state.
- The nodes are states. The links are actions. (arcs)
- state space forms a graph
- A path in the statespace is a sequence of states connected by a sequence of actions.
- Goal test - It determines whether a given state is a goal state.
- In some cases there will be a set of explicitly goal states

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- The test will examine whether the given state is the goal state.
- path cost It is a function that assigns a numeric cost to each path.
- The problem solving agent chooses a cost function that reflects its own performance measure.
- A path from initial state to goal state is called solution
- solution quality is measured by the path cost function
- An optimal solution will have the lowest path cost among all solutions.
- The process of removing details in a representation is called abstraction.

(During the journey from source to destination, the state of the world includes so many thing, like the radio, the scenery out of the window, the weather and so on... All these are left out of our description, b/c they are irrelevant to our problem)

- A rational Agent can also be called an intelligent agent. autonomous agent.

Task environments

The task environments are essentially the problems, the rational agents are the solutions.

For solving a problem rationally we use PEAS

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i.e we use performance measure the enviroment, the agent actuators and the sensors.

PSA in AI involves a systematic process where agents identify a challenge, develop strategies and execute solutions to achieve a goal.

steps:
1. problem Identification
   . Recognizing and defining the problem.
2. Formulating ~~the~~ problem
   structuring the problem in a manner where an AI agent will ~~work~~ solve. solve.
3. Strategy formulation
   Developing a plan to navigate from Initial to ~~next~~ goal state
4. Execution and monitoring
5. learning & Adaptation.

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Thumb rules are practical experience based guidelines for quick estimation or decision making

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Heuristic is a practical problem solving technique.

- A PSA in AI is a system that finds and fixes problem in its environment. rules of thumb to (find solutions quickly
- These agents use Algorithms, heuristics and search strategies to solve complex tasks. techniques
- Heuristic search use set of rules to move from one solution to another.
- It depends on experience or intuition rather than systematic analysis
- It is used to evaluate functions.
- It uses statistical inference (identifying sub parts in a graph).
- Applications
  - It is used to solve complex problems.
  - It is used in route planning.
  - It is used in computational biology.
  - It is used in robotics.
  - It finds efficient solutions faster than classical methods.

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- They are helpful in AI development of AI algorithms.

- How it work.

1. Goal formulation: Define specific objective to achieve
2. problem formulation: The agent defines the task by specifying the initial state, goal state, actions, constraints and criteria for evaluating solutions.
3. Search: The agent searches for solutions with a defined environment. (BFS) DFS)
4. evaluate/execution: The agent evaluates the options and select one or more perform the chosen sequence of actions
5. learning: Adapt & improve decisions over time based on experience

Omniscience AI: It refers to hypothetical state where an AI agent possesses complete knowledge about its environment.

It helps us to make perfect decisions based on this perfect understanding.

- No AI can truly have complete knowledge about a complex environment.

- criteria define what you want to achieve
- constraints dictate what you can and cannot do to reach the goal.

criteria refers to standards or requirements,
constraints are limitations or restrictions that limit the options available.

problem solving agents:

Agent are entities, they will take actions based on sensors.

problem solving agents stand out due to their focus on identifying and resolving issues systematically.

problem solving agents can be solved in different areas:

- Identifying challenges
- Analysing situations
- Applying strategies

Resolving issues.

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steps in problem solving

1. problem definition
2. problem analysis
3. knowledge representation
4. problem solving

A.I problems can be of mainly 3 types.

problems

Ignorable problems | Recoverable problems | Irrecoverable problems.

Here we are having same technigues

- Search Algorithm
- constraint satisfaction problems
- optimization technigues
- Machine learning
- NLP.

- To solve problem first I will check the starting state of problem or from where I have to start.
- How to reach the goal state
- what are the cases of the problem
- It should be profitable.
- problem formulation have 5 steps.

1. Initial state :
2. Actions
3. Succesor function
4. Goal state
5. path.

model result

Initial state + action + transaction action

A statespace is a mathematical concept representing all possible condition (states) a system can be in visualized form.

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Year & Sem 1.B.I Subject : AI

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Searching for solution:

Searching solution for a problem can be done by state space

- According to the state space searching solution is possible.

- Search technique that use an explicit search tree.

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- The search tree is generated by the initial state and the successor function, these together defines the state space.

statespace = Initialstate+ successorfun

- we suppose to consider search graph rather than search tree when one state having multiple paths

- The choice of which state to get expand is called search strategy.

- There are many ways to represent a node; a node is a datastructure with 5 components:

- State: The state is the state space the node corresponds.
- parent node - The node in the search tree that generated this node.
- Action: The action that was applied from the parent to generate the node.
- path cost: cost is denoted as g(n) of the path from initial state to the node as indicated by parent pointer.
- depth: the number of steps along the path from the initial state.
- In AI fringe is a datastructure that stores nodes that can be reached from the current state.
- fringe is used in search algorithms to find the best path to a goal.
- It can organised as a queue
- fringe refers to a DS that stores all the possible nodes or states that a search algorithm is currently considering.
- Simplest representation of fringe is a set of nodes.

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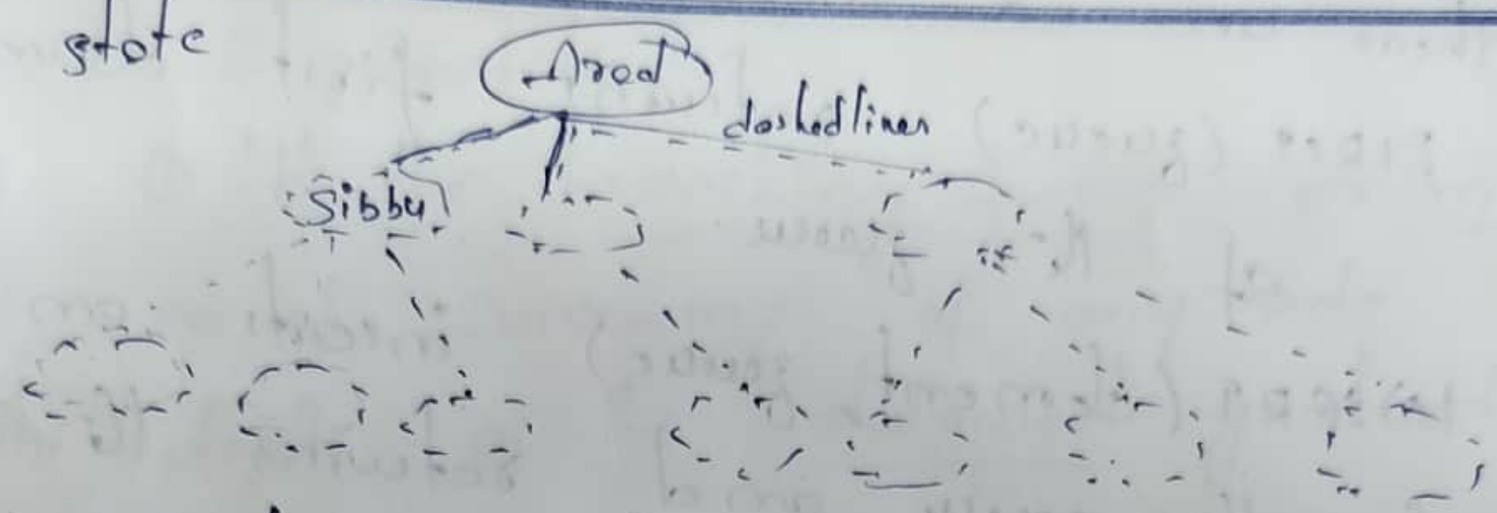
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(b) Initial state

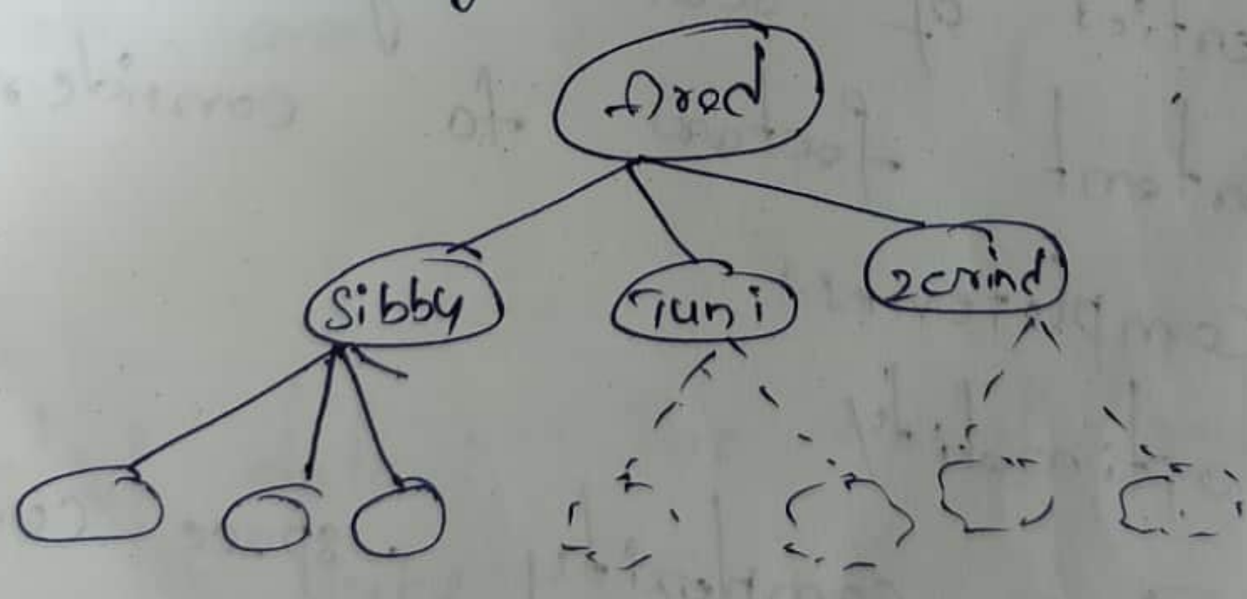


(b) After expanding Arad



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(c) After expanding sibiu



- The collection of nodes is known as fringe. It is straight forward and computationally expensive. The collection of nodes represented as queue
- The queue operations are

- MAKE-QUEUE (element, ...) creates a queue with the given elements
- EMPTY? (queue) returns true only if there are no more elements in the queue
- FIRST (queue) returns first element of the queue.
- INSERT (element, queue) inserts an element into the queue and returns the resulting queue.
- INSERT-ALL (elements, queue) inserts set of elements into the queue and returns the resulting queue.

measuring problem-solving performance

(or) properties of search algorithm.

4 important factors to consider

- completeness
- optimality
- Time complexity
- space complexity.

- In AI the graph is represented implicitly by the initial state and successor function and is frequently infinite, complexity is expressed in

terms of 3 quantities :

① 'b' the branching factor or maximum no. of successors of any node

② 'd' the depth of the shallowest goal node.

③ 'm' the maximum length of any path in the state space.

- Search cost which typically depends on the time complexity.

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- Tower of Hanoi problem

→ on one



Initial state : The no. of rods and the no. of disks.

Rods : Requires 4 rods

Disks : A set of disks, each of different size Initially these disks are stacked on Rod, in a descending order with the largest disk at the bottom.

constraints / Rules

- single move. pick and move only one disk at a time.
- Top Disk : you can only move the top disk from any of the rod.
- placement. According to the descending order
- Goal. The objective is to move the entire stack of disks from one rod to other, maintaining their size.

S.A

I

Uni

Inf

DFS BFS uniform cost search · DS greedy search A^* Graph search

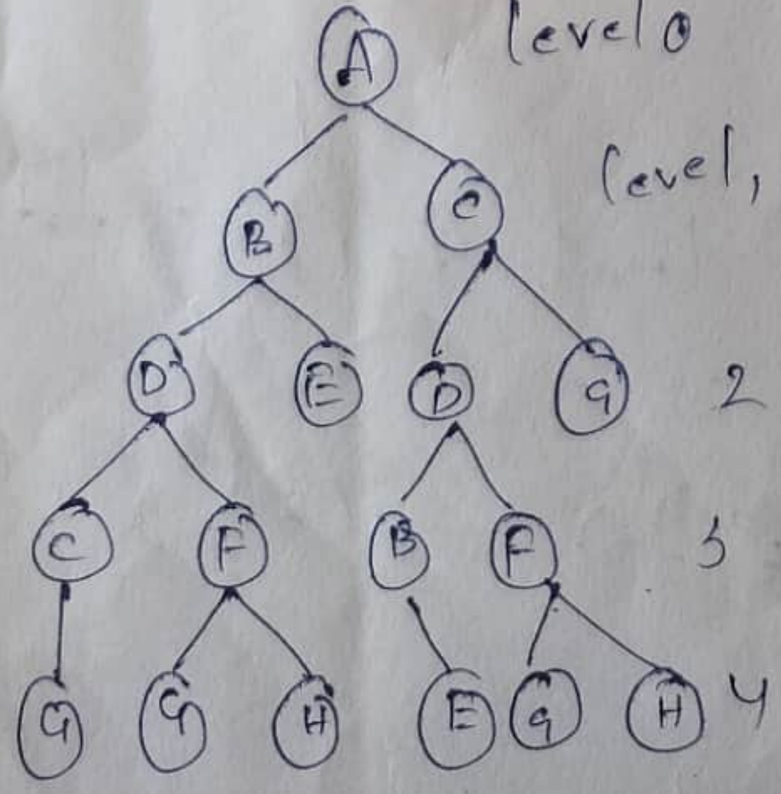
- Iterative deepening DFS
- Bidirectional.s

breadth - first search (Traversal tech

1.5

1. create a veriable called NODE-LIST & set it to the initial state.
2. until a goal state is found or NODE-LIST empty
   (if node list is empty it means goal node is not found in the path that's why empty)
   a. Remove the first element from NODE-LIST and call it as k or E. If NODE-LIST was empty then quit.
   b. for element E do the following
      i. Apply the rule to generate a new state
         (we suppose to consider the successors of E or k)
      ii, if the new state is a goal state quit and return this state.
      iii, otherwise add the new state to the end of NODE-LIST

A-C-



1. A Root node.

Rem insert

2. ← B C ← B's successors are D, G, C

3. C D G ← C's successors are D, G

4. C D E

5. D E D G D's successors are E, F



characteristics : queue (FIFO)
cost optimal
early goal test
complete

It is a level wise approach

Applications :
. shortest paths
. social n/w
. web crawling
. ML

Time complexity $O(b^d) = 2^4$
b - branch factor $2 \times 2 \times 2 \times 2$

It is an uninformed
FIFO
shallowest node consideration.

$b^0 + b^1 + b^2 + b^3 + \ldots b^d$
$1 + 2 + 2^2 + 2^3 + 2^4$
$3 + 4 + 8 = 15$

$$\frac{b^{d+1} - 1}{b - 1}$$

$$\frac{b^{3+1} - 1}{2 - 1} = 2^4 - 1$$
$$2 - 1 = 15$$

5. ~~C~~ F E D G C F

6. E have no successor node

6. D G C F
D's successor nodes are B E F

7. G C F B F

8. G is the goal node.
A - C - G is the paths to reach the goal.

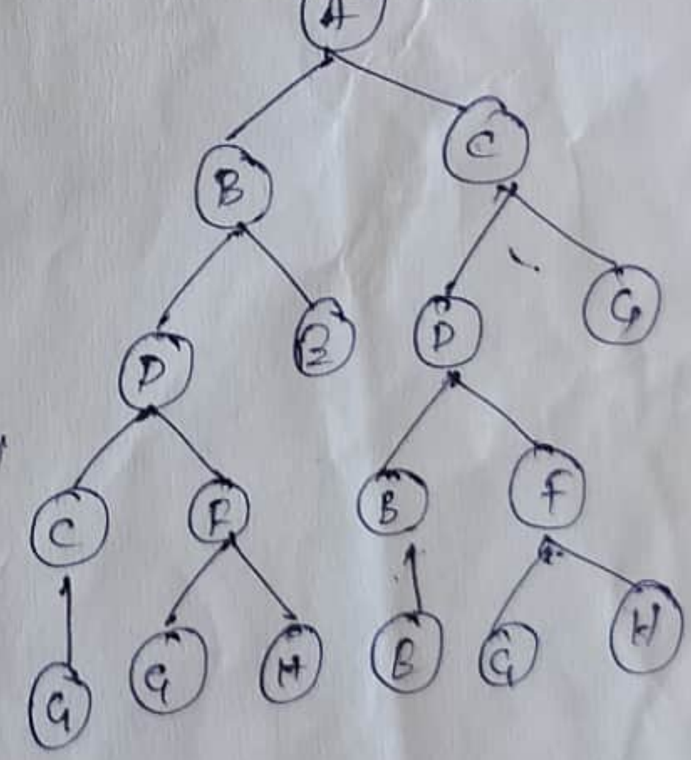
All the nodes are
generating $b^{d+1} - b$ nodes at level $d+1$; so total no. of nodes generated is

$$b + b^2 + b^3 + \ldots b^d + (b^{d+1} - b) = O(b^{d+1})$$

while designing I.A for complex systems AI engineers use the PEAS framework. (8:-7)

DFS:

1. If the initial state is a goal state, quit & return success.
2. else do the following until success or failure is signaled (no node)
   a. Generate a successor, call of the initial state. If there are no successors, failure.
   b. Call DFS with E as the initial state
   c. If success is returned, signal success else continue in the loop.



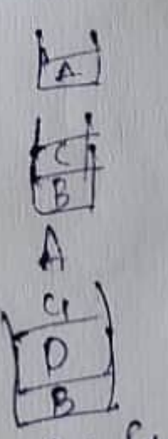
NODE list : A

successors of A is BC

NODE list : BC

B is not the goal node

successors of B are D & G



NODE list DEC

D is not the goal node

so its successors of D are C & F

NODE LIST CFEC

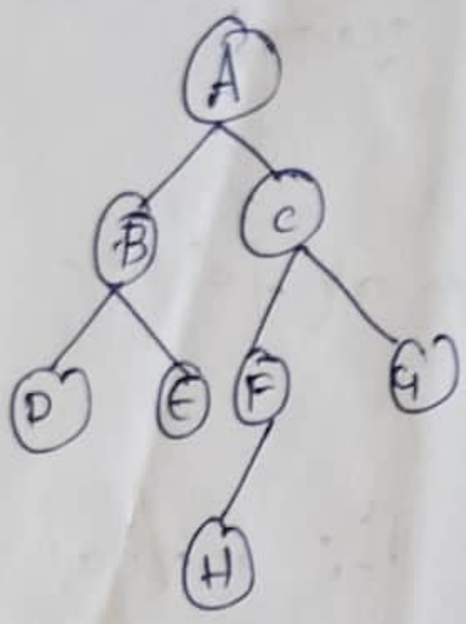
e is not the goal node

successor of e is G

So the path is A-B-D-C-G

Advantages:

. It is used for bolt trees & graphs.



Time. compler
o(v+e)

Spere
o(v)

stack (LIFO)
Deepest node
Incomplete

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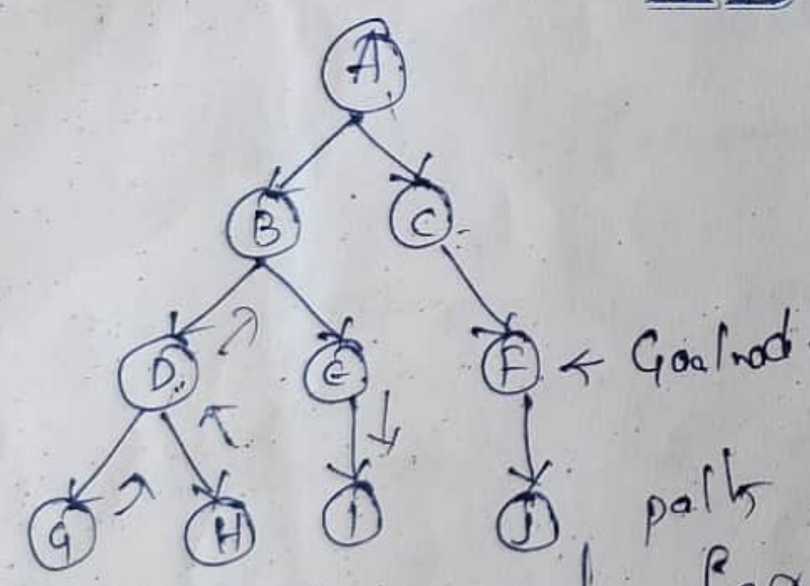
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Subject : AI (1.6)
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Depth limited search Algorithm: (DLS) (DLDFS)

DLS is similar to DFS with a predetermined limit.

- DLS will solve the problem of DFS and the search of it will solve the drawback of the infinite path in DFS.
- The Depth limit will consider as if has no successor nodes.

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The traversal path DFS is A B D G H E I C F

DFS: A B D G H E I C F

Depth limit = 2

Depth limit = 2

DLS:

A B D G E C F

Time complexity $O(b^l)$ or

$O(b^d)$

space complexity $O(b \times l)$

completeness: is complete only if the solution lies with in the set depth limit.

If the goal state is deeper than the limit it is incomplete, in that scenario

optimality:

It is not optimal b/c it prioritizes searching within the depth limit, by means ignoring the solution in the nearest level.

- Uniform cost search is an algorithm used in AI to find the least expensive path in a graph. every time we can sort
Initialize cost is 0.
- ucs explores nodes in a graph based on the cost of reaching them.
- It uses a priority queue to expand nodes with the lowest path cost first.

FR-SRIT

- ucs is used when the edges of a graph have different weights.

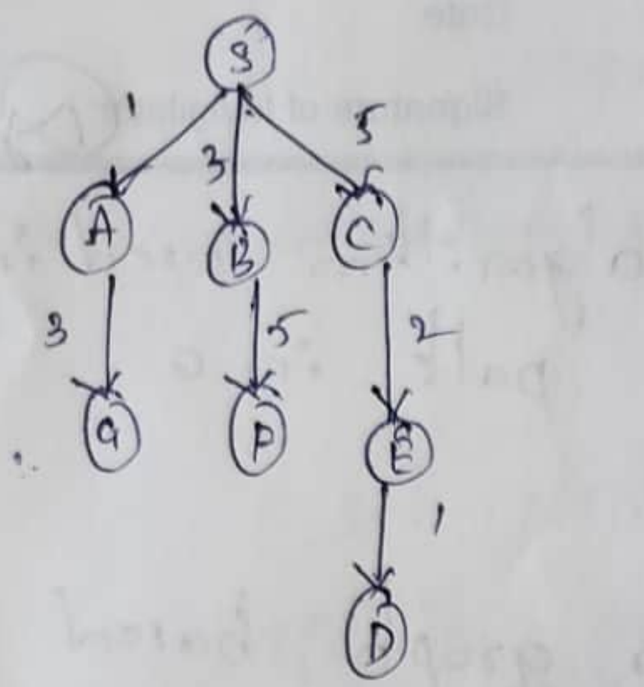
. Insert the root node into the priority queue.

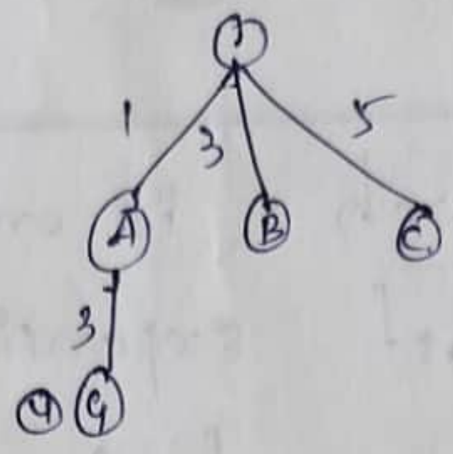
. Remove the element with the highest priority

. If the removed node is the goal node
- print total cost and stop the algorithm

else enqueue all the children of the current node to the priority queue with their cummulative cost from the root as priority and the current node to the visited list.

- which node is having low cost that means it is having higher priority.

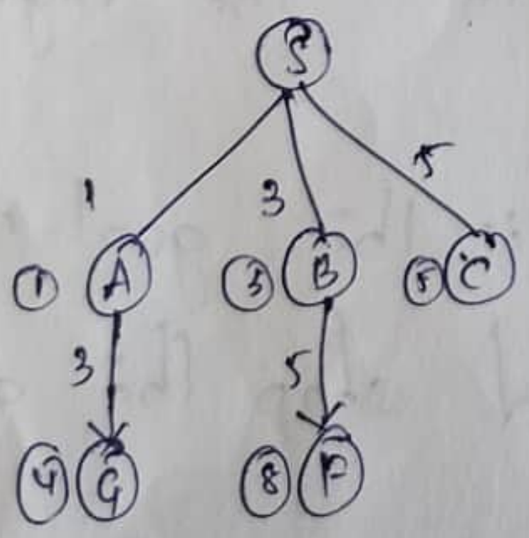




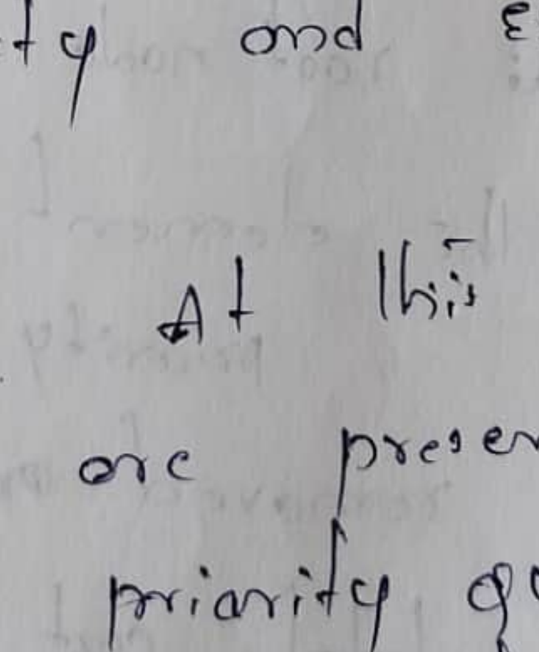
in all A, B, C A cost is less means having highest priority exploring A.

- we will find which node is having higher priority and we will explore A that node removed &

- compare the remaining leftover nodes G, B, C in all there B is having lowest cost, so. consider B, it it having highest priority and explore it.



At this stage G, F, and C are present in it priority queue out of these 3 G it having minimum cost no nodes for G so we stop considering G.



next remaining C and F, C it having minimum cost

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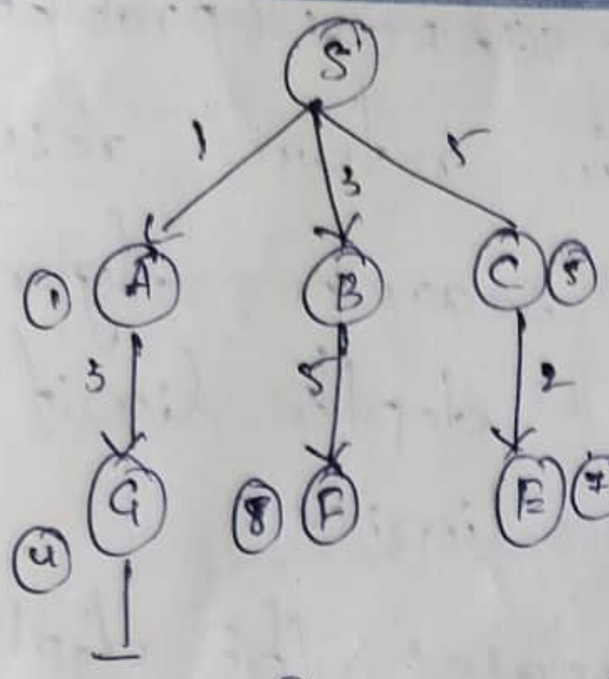
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Subject : AI Lab

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Date :

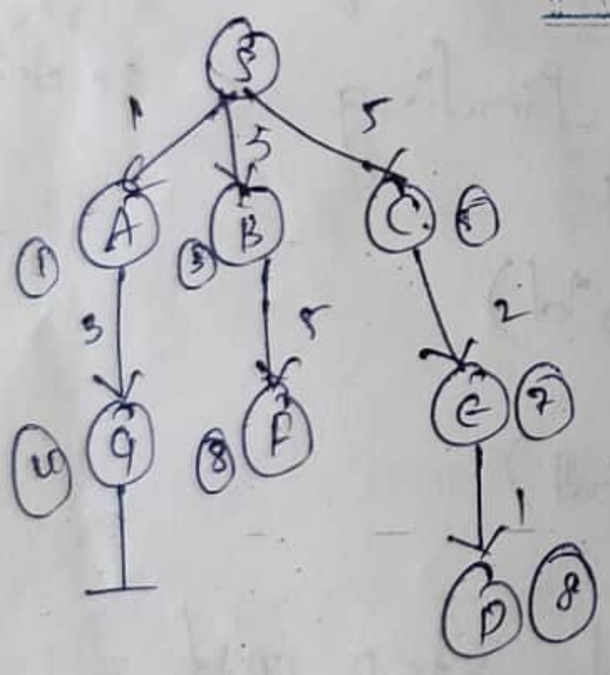
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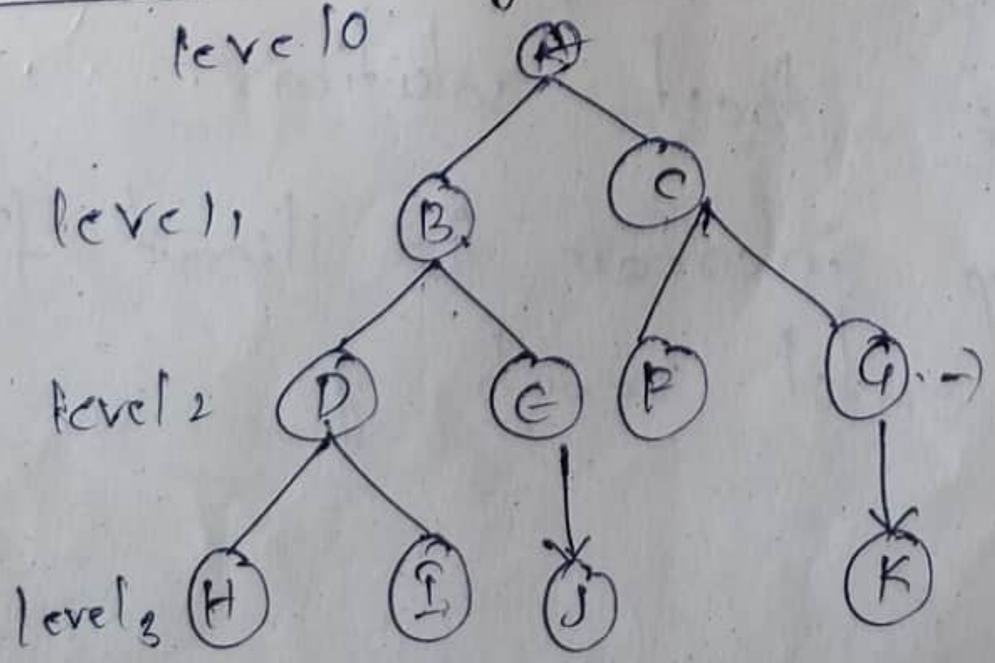
considering c in IIr priority queue and exploring it childrem.

→ from B and F, E is having the highest priority

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Iterative Deepening DFS



Iteration 0 : A

Iteration 1 : A → B → C

Iteration 2 :

A → B → D → E → C → F → G

Iteration 3 :

A → B → D → H → I → E → J → C → F → G → [illegible]

- function ITERATIVE-DEEPENING-SEARCH (problem) returns a solution, or fa
inputs: problem
for depth ← 0 to ∞ do
result ← DEPTH LIMITED-SEARCH (problem, depth)
if result ≠ cutoff then return result

The iterative deepening search algorithm which repeatedly applies depth limit search with increasing limits.

- Iterative deepening is increases the depth limit to ensure completeness.
- It is still being memory efficient compared to other strategies like BFS.

completeness: completes finding a solution. increasing depths, (yes
Time complexity is $O(b^d)$

space complexity: $O(bd)$

optimality: yes if step cost = 1

→ combines the benefits of BFS and DFS to find the best solution.

→ If gradually increase the limit from 0, 1, 2 .. until reach g

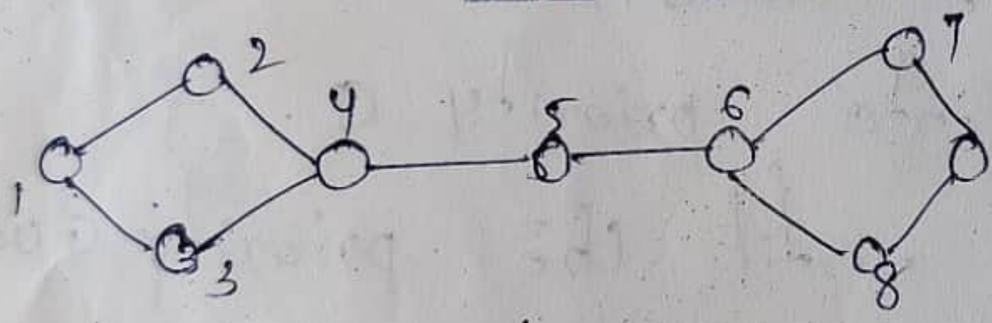
Bi-directional search algorithm: It is an uniformed search technique

→ Searching is done in two directions. forward

→ simultaneous search from on inital node to goal and backward from goal to initial stopping when two meet.

Sometimes lead to find solution more quickly.

EB-SBIT



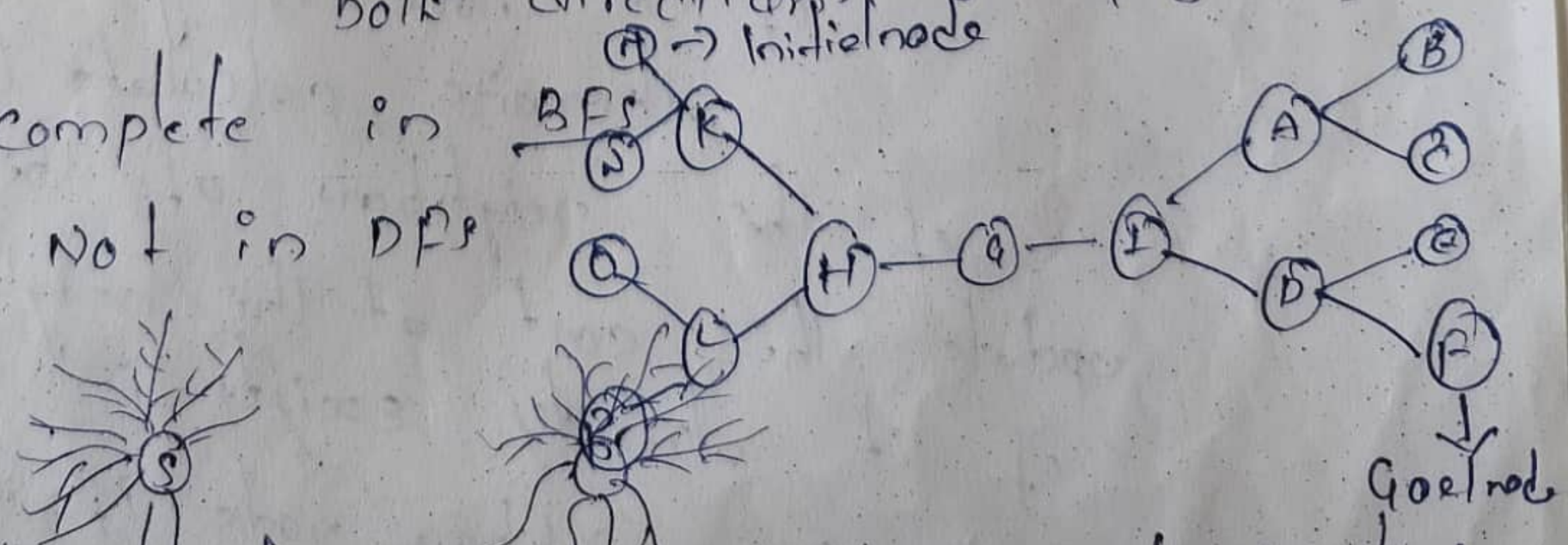
- Time complexity: worst case $O(b^d)$ same as DFS and BFS.

$2(b^{d/2})$

Both directions, $b^{d/2} + b^{d/2} =$

- complete in BFS

Not in DFS



parallely searching is done from 16 initial state and gain goal state.

complete : yes (if any single goal, otherwise difficult)

Time complexity : $O(b^{d/2})$ (better when compared to other algorithms)

space : $O(b^{d/2})$

optimal : yes.

UCS : finds the path with the lowest total cost b/w 2 nodes in a graph/tree.

In UCS we can use priority queue.

It is a variant of Dijkstra's algorithm.

1. first initialize a priority queue
2. set the initialized cost to 0 and push it into priority q.
3. Repeat until the priority queue is empty.
   - which node has lowest cost to remove dapmet.
   - If the node is goalnode - end otherwise calculate the path cost of the neighbors of the current node.
   - update the cost of the neighbor node if it already exists in the pq (priority q) push the neighbor node if it doesn't exist in the pq. (priority q

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Informed Search strategies : It use domain specific knowledge, called heuristics, to guide search. A* It is used to find solutions very quickly than uninformed.

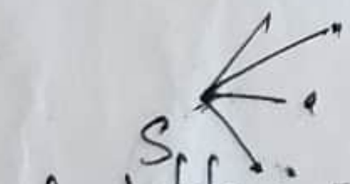
→ Heuristic in AI (Rule of thumb) { what, why, How }

Bfs : A Non polynomial problem can be converted into polynomial type based on the heuristic

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To reduce the distance we suppose to consider euclidean distance (K-Mean KNN)

$d((x,y),(a,b)) = \sqrt{(x-a)^2+(y-b)^2}$ $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

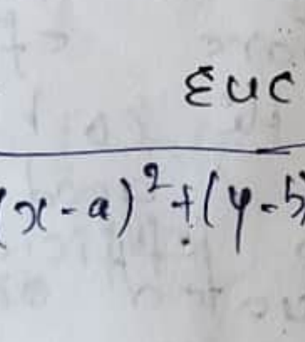


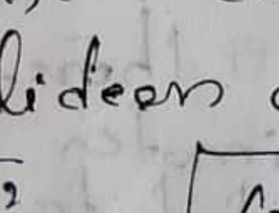
It is called straight line distance. b/w 2 points or data points or vectors

Manhattan distance : find out the positions distance b/w 2 points by summing the absolute differences of their coordinates, $d = |x_1 - x_2| + |y_1 - y_2|$ for 2D

N. D (L1norm / Taxicab distance) calculate,

Grid Based movem (8 puzzle problem)





0 + 1 + 1 + 2 + 0 + 2 + 2 + 0

we suppose to get many distances that by using manhattan which is going to give minimum that is going to get selected.

we are calculating the heuristic values. Based on that we will consider the nodes.

- The heuristic will give the good solution but not the optimal solution.

Best first search:

Informed search (Heuristic search) / pure heuristic search

- It is also called intelligent search
- It uses information about the problem to guide the search,
- usually guesses the distance to a goal state and therefore efficient search.

H.F is a function that estimates the cost from the current state_id(node) to the goal state. A good heuristic node can be evaluated or expended by using improves efficiency heuristic function.

- The evaluation function is f(n).

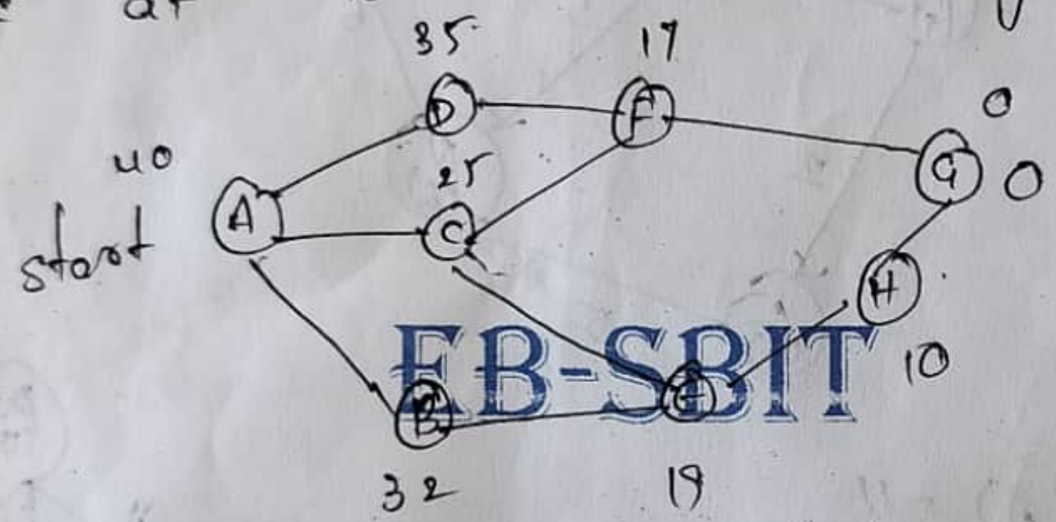
Greedy Best First search in AI

Greedy best-first search tries to expand the node that is closest to the goal, the grounds that this is likely to lead to a solution quickly.

- It evaluates nodes by using the heuristic function, i.e evaluation function

$f(n) = h(n)$ →

- $h(n)$ is the estimated cost from the state at node n to a goal state.



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$f(n) = h(n)$

$h(n)$ = estimated cost ~~function~~ from the state at node n to a goal state.

$f(B) = h(B) = 32$

$f(C) = h(C) = 25$

$f(D) = h(D) = 35$ — (1)

$f(F) = h(F) = 17$

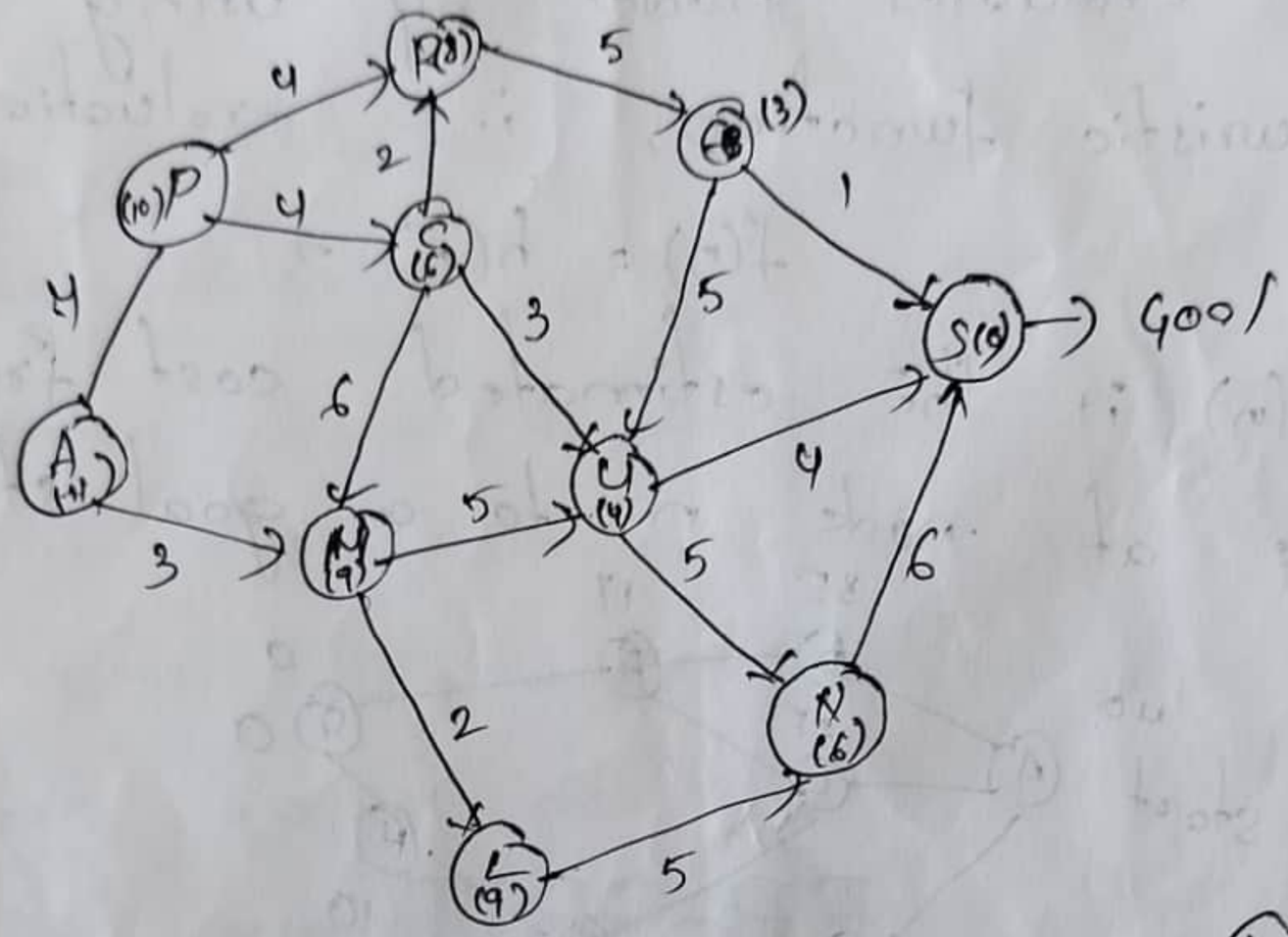
$f(E) = h(E) = 19$

$f(D) = h(D) = 35$

$f(G) = h(G) = 0$

$A \rightarrow C \rightarrow F \rightarrow G$

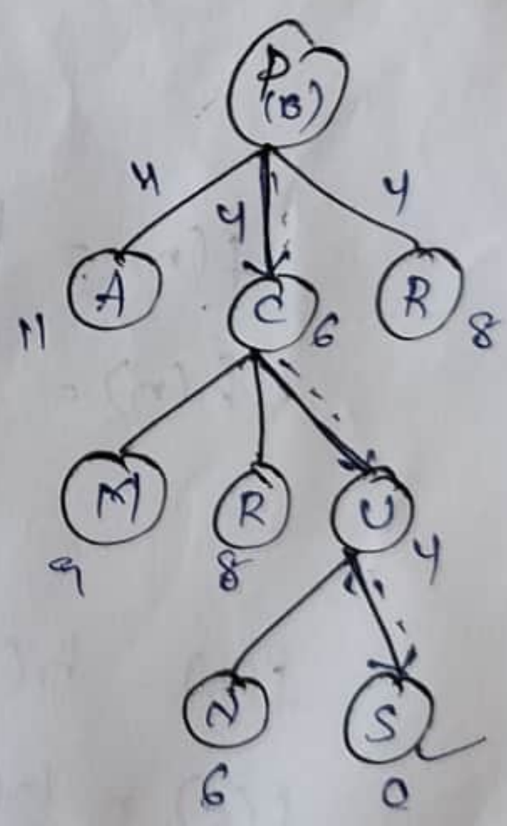
GBFS tries to expand the node that is closest to the goal, on the grounds i.e likely to lead to a solution quickly.



$f(A) = h(A) = 11$

$f(C) = h(C) = 6$

$f(R) = h(R) = 8$



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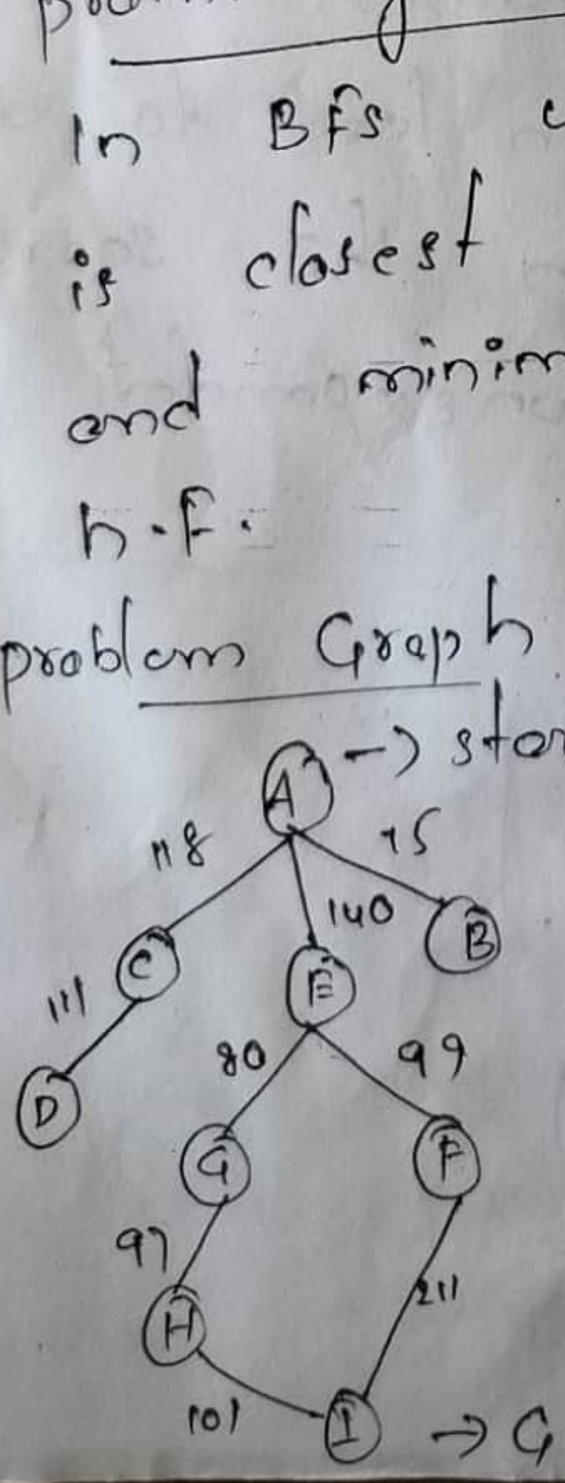
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Greedy best-first search

- GBFS algorithm always selects the path which appears best at that moment.
- It is the combination of DFS and BFS Breadth first search algorithms.
- It uses heuristic function and search.
- Using Best first search at each step we can choose the most promising node.

In BFS we expand the node which is closest to the goal node and minimum cost is estimated by h-f:

problem Graph

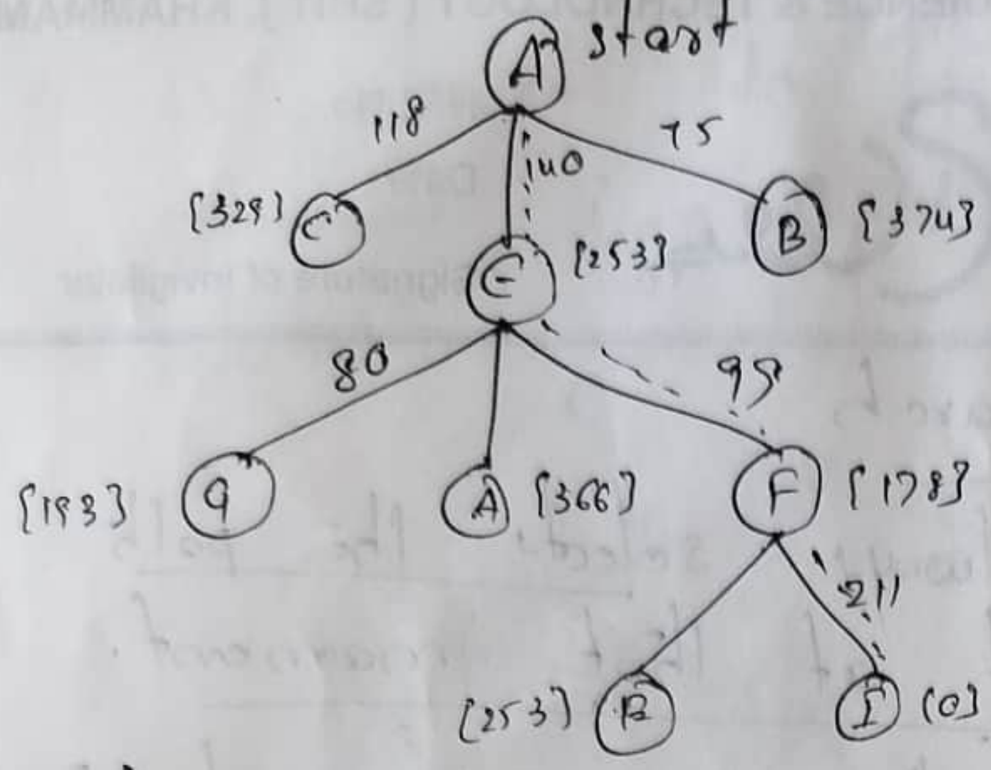


Heuristic Table

state	heuristic(n)
A	366
B	374
C	329
D	244
E	253
F	178
G	193
H	98
I	0

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$f(n) = h(n) =$ straight line distance function

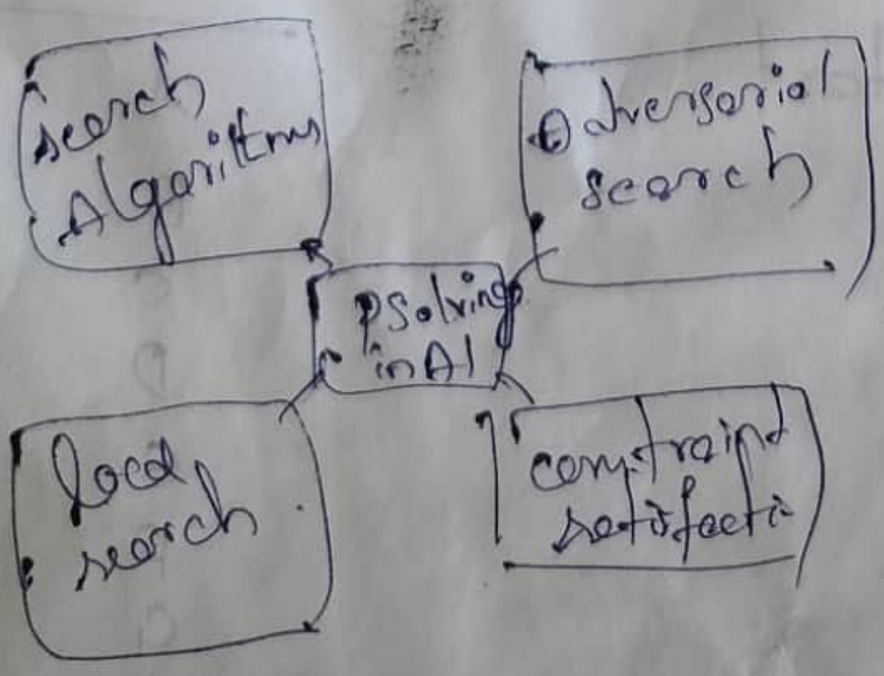


path cost = A - E → F - I = 253 + 178 + 0 = 431

dist(A-E-F-I) = 140 + 99 + 211 = 450

- GBFS can start down an infinite path and never return to try other possibilities.
- Because of its greedines the search makes choices that can lead to dead end, one back up in the search tree tree to the deepest unexpanded node.

General



3/3/25 (1)

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- BFS combines the advantages of DFS and BFS into a single method.
- At each step of the BFS search procen, we select the most promising node.
- It is done by applying heuristic function to each of them.
- we then expand the chosen node by using the rules to generate its successor

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- we need To implement the BFS
- we use two lists of nodes.

OPEN - nodes that have been generated and have had the heuristic function applied to them, but which have not yet been examined (i.e had their successors generated).

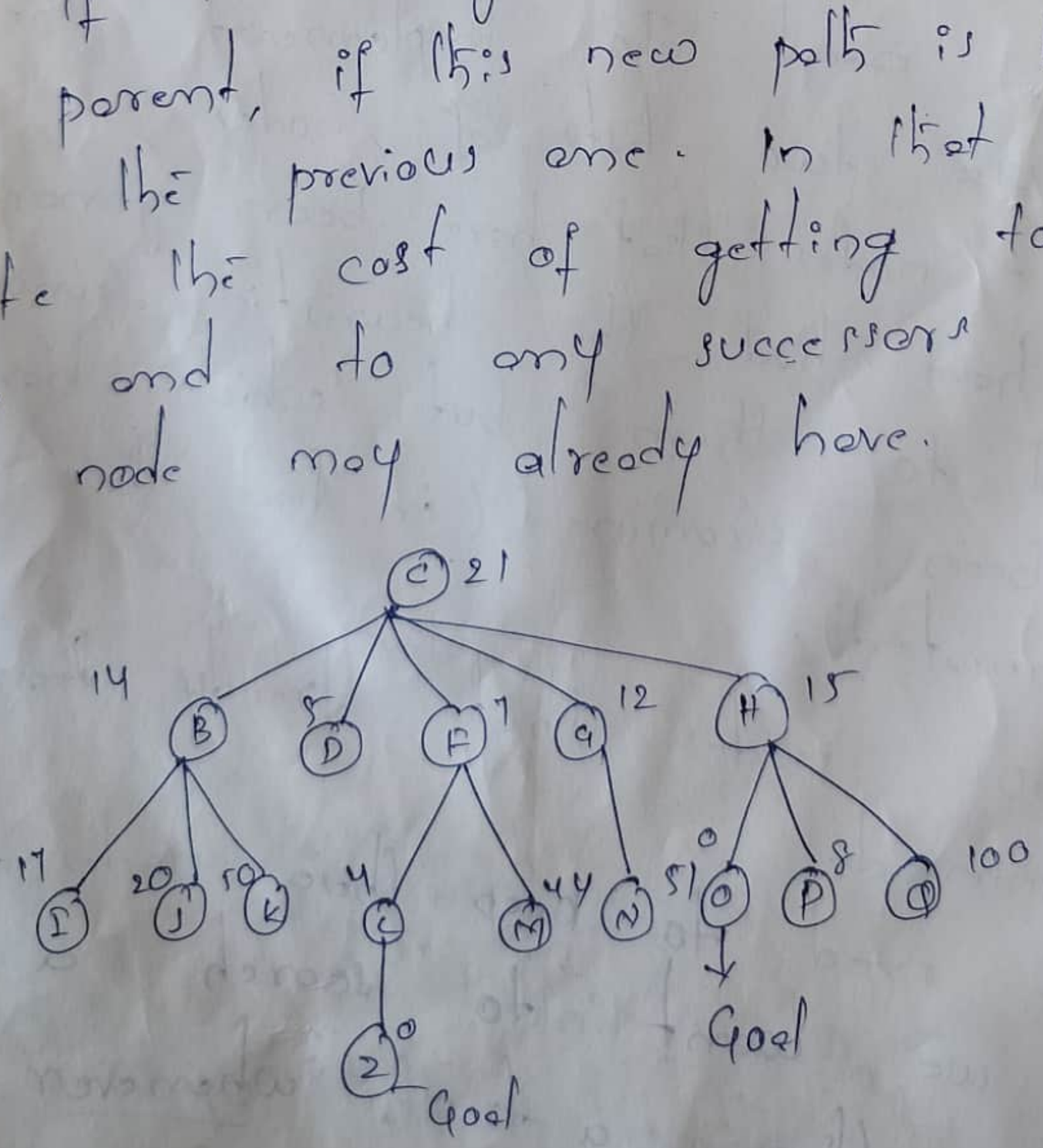
CLOSED - nodes that have already been examined.

we need to keep these nodes in memory if we want to search a graph rather than a tree, whenever a node

is generated, we need to check whether it has been generated before.

Algorithm.

1. start with open containing just the initial state.
2. Until a goal is found or there are no nodes left on open do.
   a. pick the best node on open.
   b. Generate its successor
   c. for each successor do.
      i, if it has not been generated before evaluate it, add it to open and record its parent.
      ii, if it has generated before change the parent, if this new path is better than the previous one. In that case update the cost of getting to this node and to any successors that this node may already have.



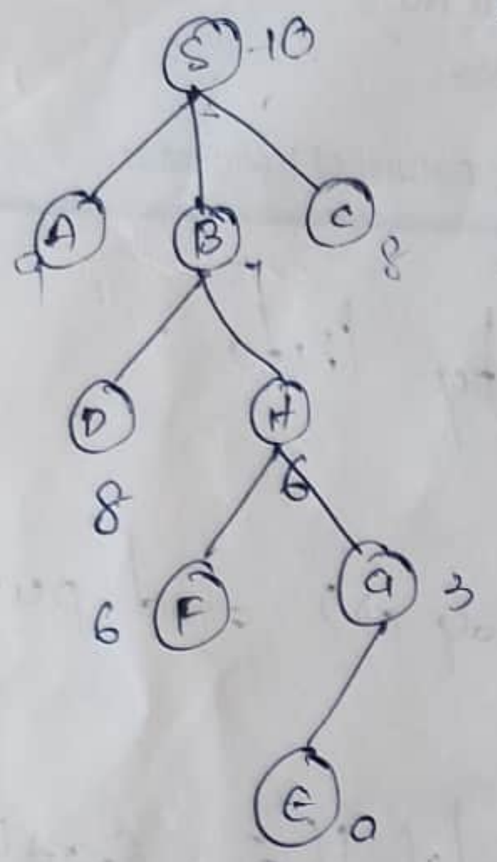
G → 4 = 0

Best - first search.

- In this we will create 2 empty lists : OPEN and CLOSED
- start from the initial node (say N) and put it in the ordered OPEN list.
- Repeat the next step, until the GOAL node is reached or OPEN list is empty.
1. select the best node (say N) in the OPEN list. Take the information of the parent node.
2. If N is a GOAL node, then exit the loop returning True. The solution can be found by backtracking the path.
3. If N is not the GOAL node, expand node N to generate the immediate next nodes linked to node N and add all those to the OPEN list

EB-SBIT

4. Reorder the notes in the OPEN list in ascending order.



open		closed	
Node	H(n)	Node	parent
S	10	S	S

open		closed	
Node	H(n)	Node	parent
A	9	S	
B	7		
C	8		

nodes arrange them in the ascending order

open		closed	
Node	H(n)	Node	parent
'B' is the best node and put into the closed			
B	7	S	
C	8		
A	9		

B is the best node and consider the node and place into the closed DS.

open		closed	
Node	H(n)	Node	parent
C	8	S	B
A	9	B	S

The successor nodes of 'B' are D and H' add D and H to the open DS

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open		closed	
Node	H(n)	Node	parent
C	8	S	
A	9	B	S
D	8		
H	6		

re arrange them in ascending order

open		closed	
Node	H(n)	Node	parent
H	6	S	
C	8	B	S
D	8		
A	9		

H is holding the best value consider it is not the goal. place it into the closed list.

open		closed	
Node	H(n)	Node	parent
C	8	S	
D	8	B	S
A	9	H	B

H is having 2 successor nodes F and G add to the node Datastructure

open		closed	
Node	H(n)	Node	parent
F	6	S	
G	3	B	S
C	8	H	B
D	8		
A	9		

ER-SBIT

after arranging in ascending order G is b
best so place it in the closed Datastructe

open		closed	
node	H(n)	node	parent
G	3	S	
F	6	B	S
C	8	H	B
D	8		
A	9		

G is the best place it into the closed Data structure

open		closed	
node	H(n)	node	parent
F	6	S	
C	8	B	S
D	8	H	B
A	9	G	H

for G the succesor node is E place it in

the open DS.

open		closed	
node	H(n)	node	parent
E	0	S	
F	6	B	S.
C	8	H	B
D	8	G	H
A	9	E	G

we reached the final goal node E.

so the path $S \to B \to H \to G \to E$.

Greedy Best first search:

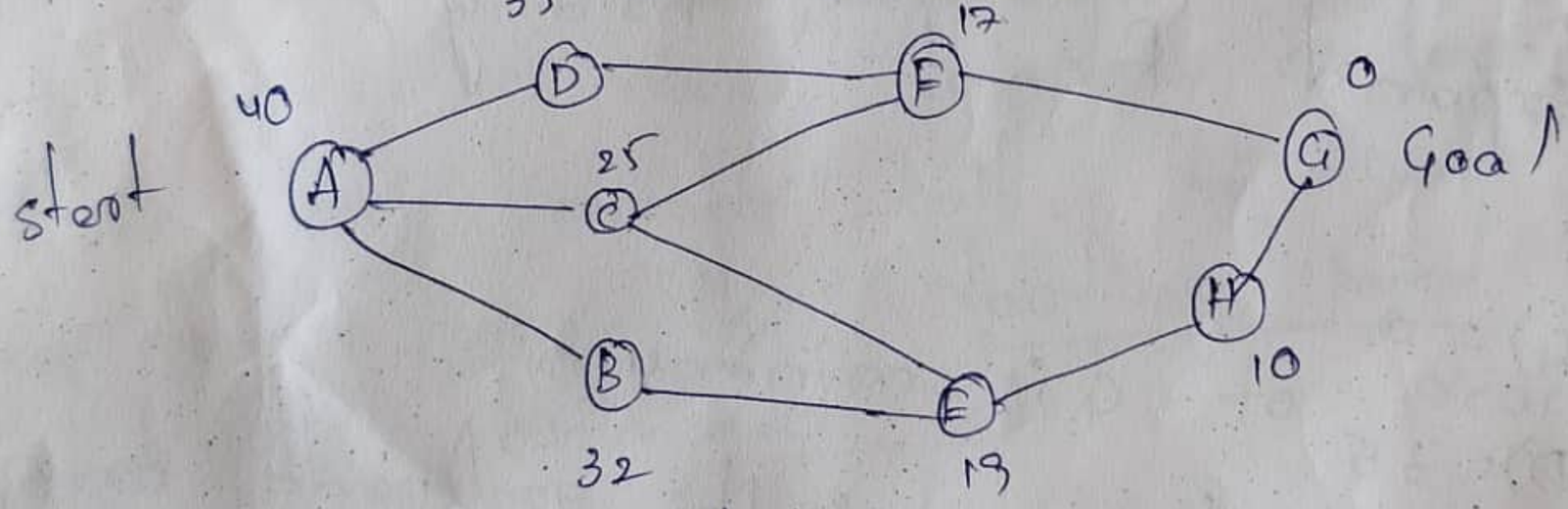
GBFS tries to expand the node i.e close to the goal node.

If leads to solution quickly.

- GBFS evaluates nodes by using the heuristic function, i.e $f(n) = h(n)$

$h(n)$ is the estimated cost from the state at node n to goal node.

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- A is the initial node, G is the goal node
- The values that are holding on each node is the heuristic value i.e $h(n)$ or estimated cost value of the node.
- 40 is the estimated cost from node A to reach the goal node G.

- The initial state is A, so we will start from initial state.
- verifying that A is the goal node or not.

$f(B) = h(B) = 32$

$f(C) = h(C) = 25$

$f(D) = h(D) = 35$

C is having the minimum value so expand C to reach the goal node.

E is having two extensions F and G if it can be represented as

$f(F) = h(F) = 17$ → minimum.

$f(E) = h(E) = 19$

So Expand 'F', F is holding minimum value.

$F(G) = 0$

$F(D) = 35$

G is minimum

verify whether G is the goal node or not.

so 'G' is the goal node.

so the path is $A \rightarrow C \rightarrow F \rightarrow G$

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- Best FS and Greedy BFS both use heuristic function to guide their search.

→ GBFS consider the estimated cost to reach the goal from the current node at each step.

→ BFS can consider the total estimated cost to reach to the goal.

EB-SBIT

→ BFS uses a more complex evaluation function that can include both the cost to reach a node from the starting point (g(n)) and the estimated cost to reach the goal from that node (h(n)).

→ GBFS: only uses the estimated cost to reach the goal from the current node h(n) as its evaluation function.

→ BFS uses the past path cost for optimality but potentially less.

→

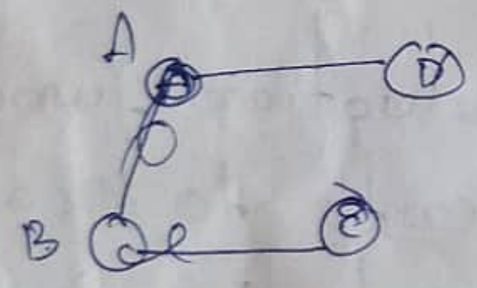
Best first search (Informed, Heuristic)

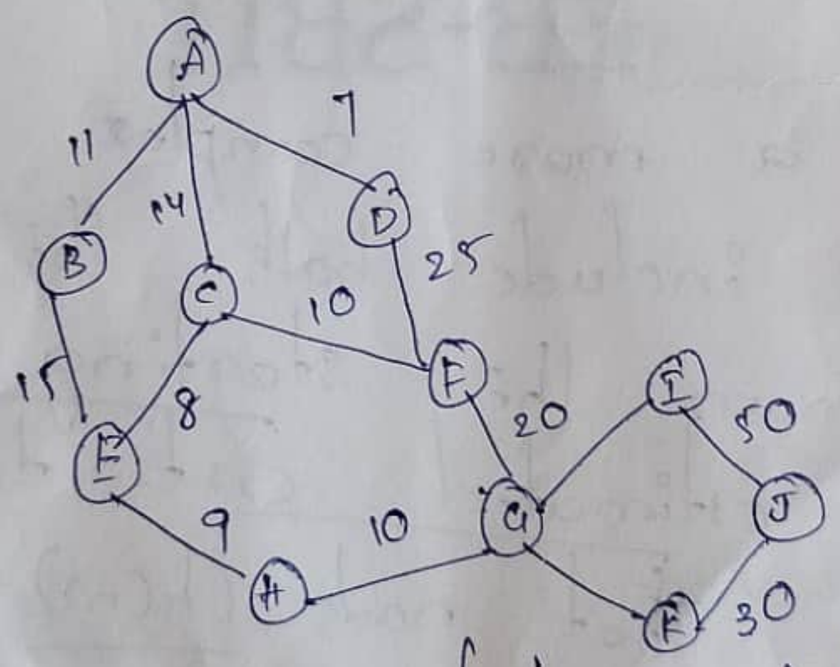
Algorithm:

- let open be a priority queue containing initial state.

loop
  if open is empty return failure
    Node ← Remove - first (open)
    if node is a Goal
  then return the path from initial to Node.
else
  generate all successors of node and
  put the newly generated Node
  into open according to their f values

END LOOP.





G is the goal node.

straigh line distance euclidean distance

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

A → G = 40
B → G = 32
C → G = 25
D → G = 35
E → G = 19
F → G = 17
H → G = 10
G → G = 0

1.9

A* Algorithm: Informed searching Algorithm.

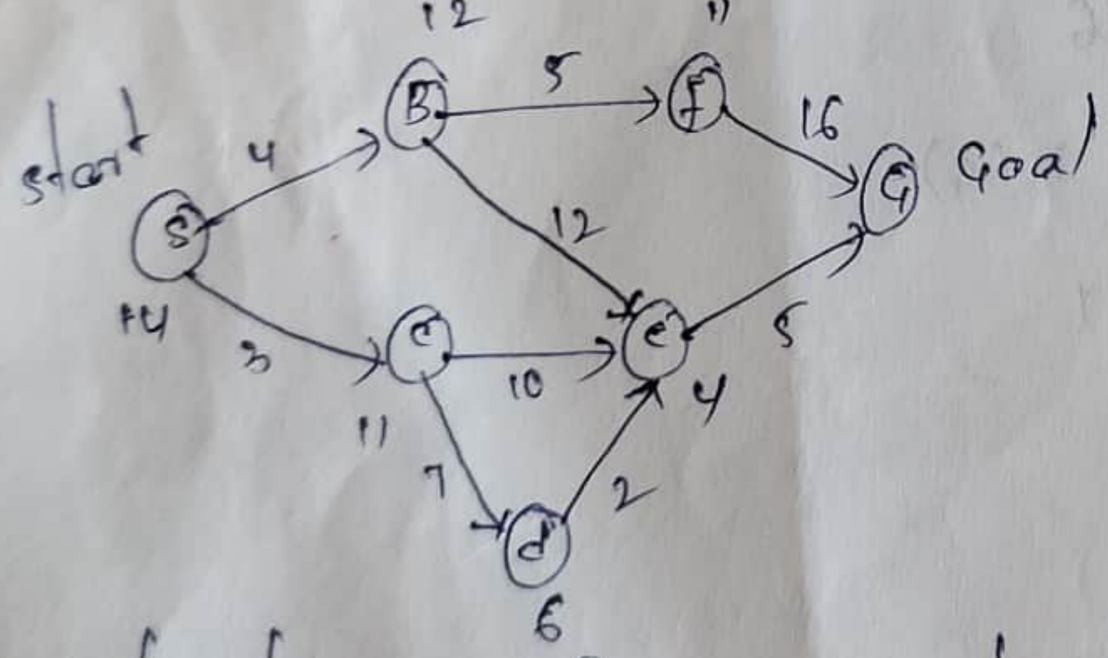
$f(N) = g(N) + h(N)$

- Having same knowledge and moving according to the heuristic
- Based on the above equation the total A* problem is going to get solved.

g(n) - is the actual cost from start node to n.

h(n) - is the estimation cost from n to Goal node. (means heuristic)

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- starting from node s it is having two ways B and C we have to select which is minimum, we have to calculate by using above function

$S \to B$ $S \to C$

A* is an admissible Algorithm.

admissible means if suppose to give optimal solution.

$f(s) = 0 + 14 = 14$

$S \to B$ and $S \to C$ $S \to C + h(C)$

$S \to B + h(B)$

$f(B) = 4 + 12$ $f(C) = 3 + 11 = 14$

$= 16$ 14 is minimum

$SC \to e$ $S \to C \to d$

~~$SC \to d$~~

$S \to C \to d + h(d)$

$S \to C \to e + h(e)$. $3 + 7 + 6 = 16$

$3 + 10 + 4$

17

16 is minimum.

$S \to C \to d \to c + h(c)$

$3 + 7 + 6 + 2 + 4 = 16$

$S \to C \to d \to e \to G + h(G)$

$3 + 7 + 2 + 5 + 0 = 17$

A* Search Algorithm example 8 puzzle problem.

Find the most cost effective path to reach the final state from initial state using A* Al.

2	8	3
1	6	4
7		5

Initial state

1	2	3
8		4
7	6	5

final state.

- Aim is to find the most cost-effective path to reach the final state from initial state

$$f(n) = g(n) + h(n)$$

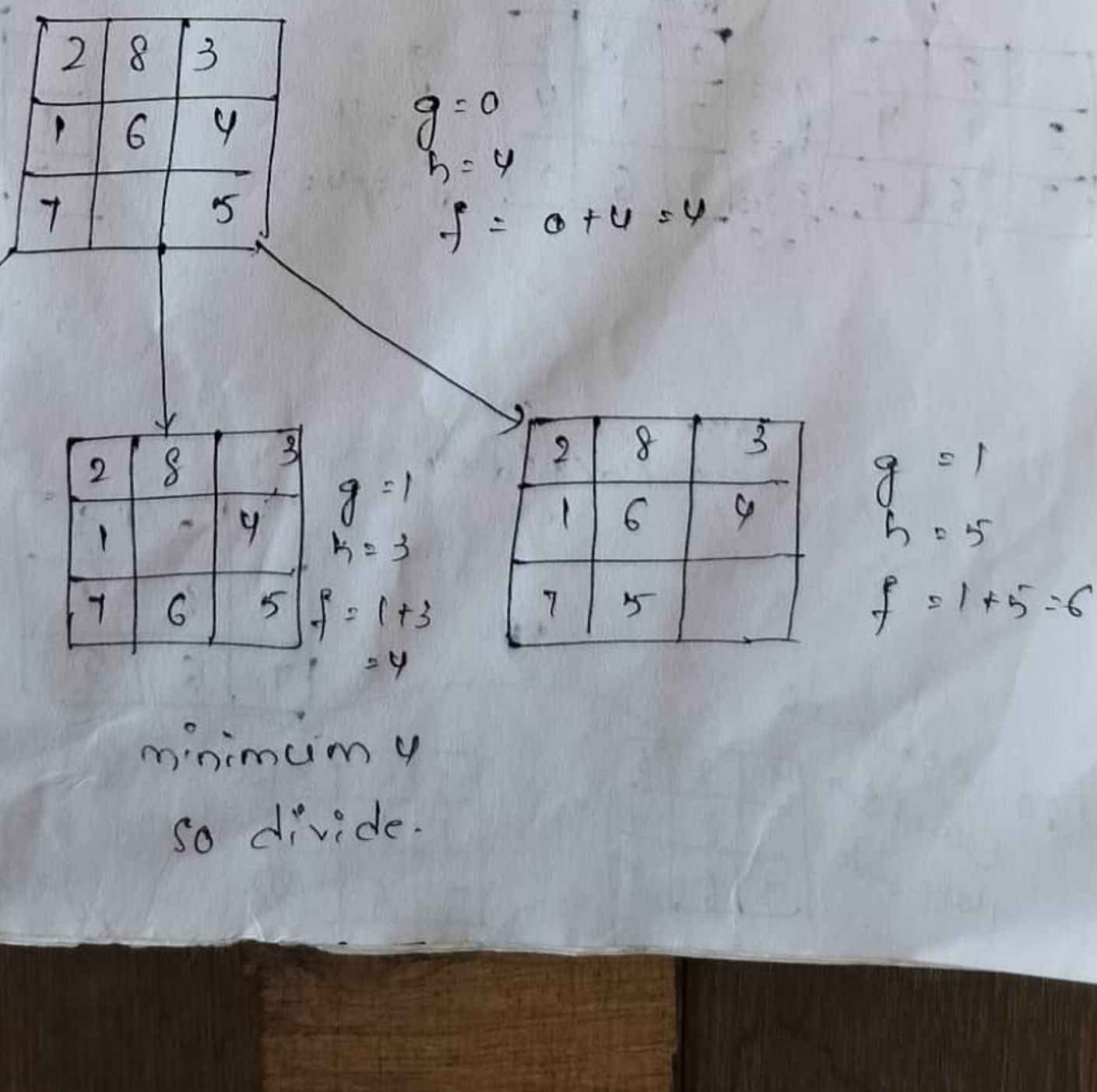
$g(n)$ = Depth of node

$h(n)$ = number of misplaced tiles.

$g = 0$ $h = 4$ are misplaced

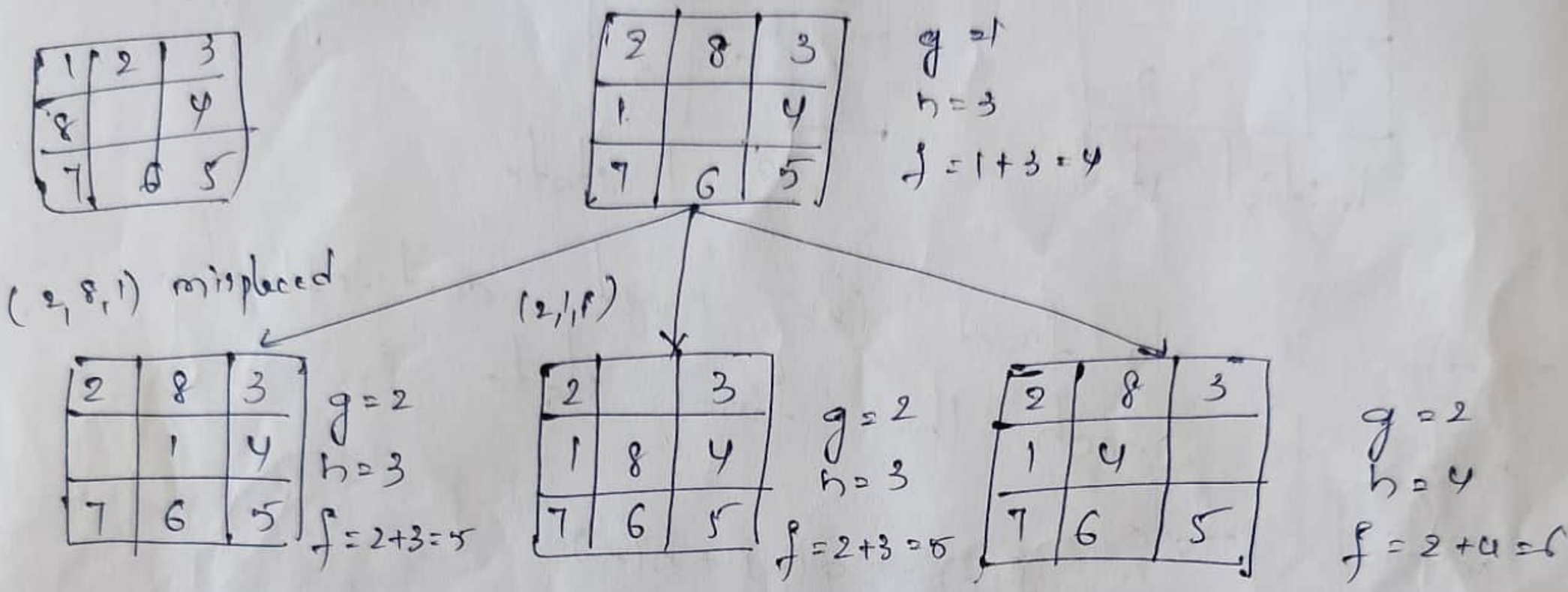
2, 8, 1, 6

Comparing with the final state how many tiles are misplaced i.e heuristic value.



minimum 4 so divide.

- Comparing with the above the figure we will see how the tiles are replaced with the Goal state, that will specify the heuristic value.



Consider

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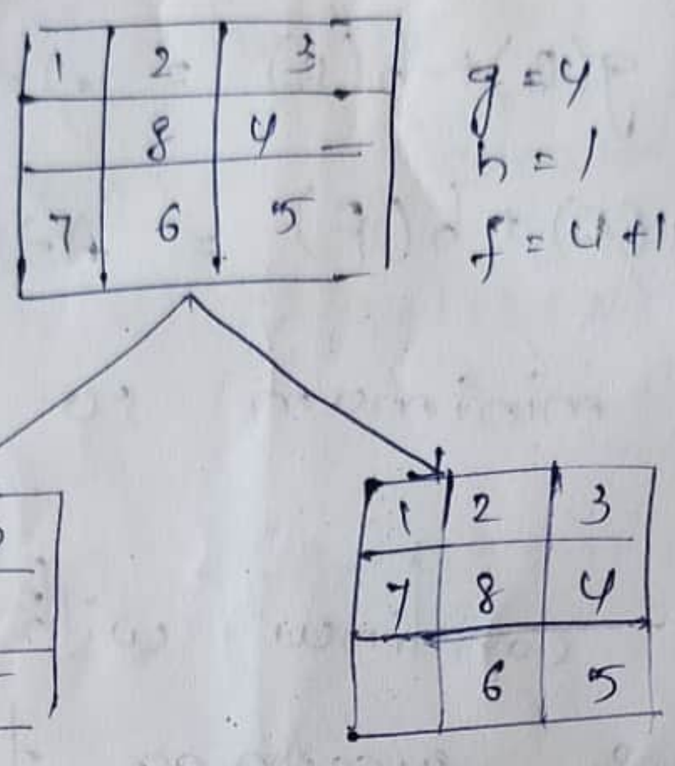
Year & Sem

Subject : AI

H.T. No. :

Date :

Signature of Invigilator

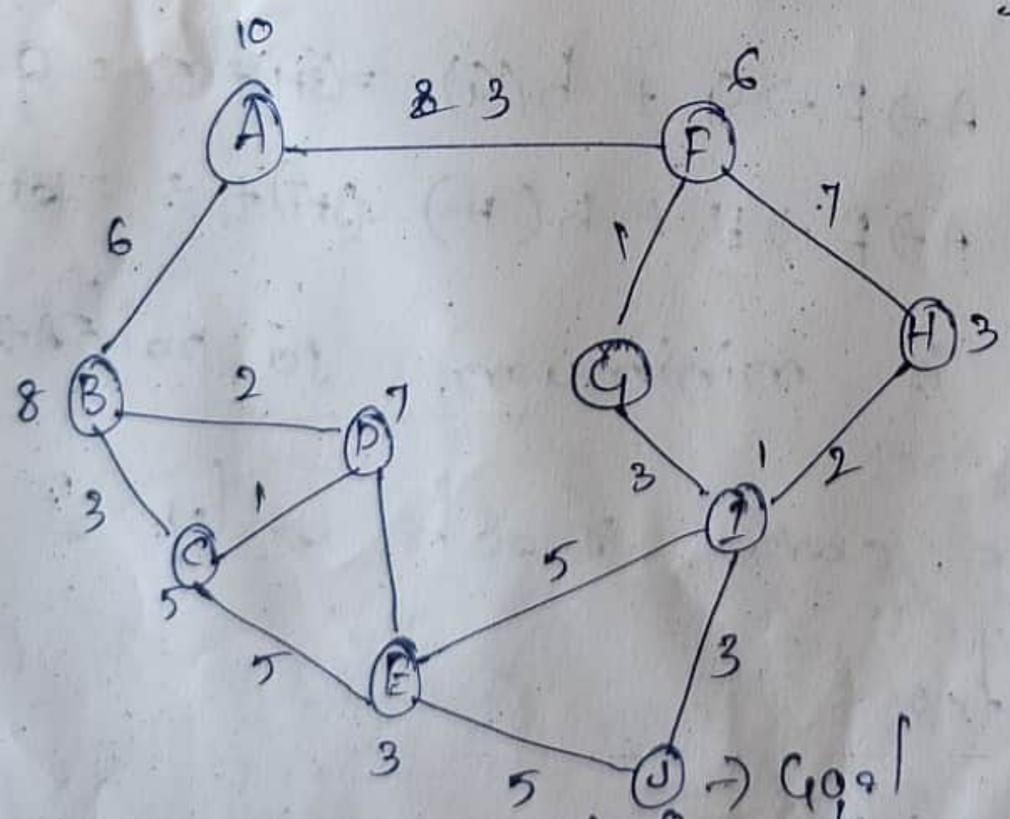


final state

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Example 2

2. Solve the graph using A* Algorithm.



Find the most cost effective path to reach from the initial state A to final state 'J' using A* algorithm.

step 1: we start with node A.

$A \to B$ and $A \to F$

A* algorithm calculate

$$f(n) = g(n) + h(n)$$

$f(B) = g(B) + h(B) = A \to B + h(B) = 6 + 8 = 14$

$f(F) = g(F) + h(F) = A \to F + h(F) = 3 + 6 = 9$

9 is minimum so path $A \to F$

step 2: we continue with node F

F is successor of 'G' and 'H' nodes

$F \to G$ and $F \to H$

A* calculates.

$f(G) = g(G) + h(G) = A \to F \to G + h(G) = (3+1) + 5 = 9$

$f(H) = g(H) + h(H) = A \to F \to H + h(H) = (3+7) + 3 = 13$

9 is minimum so path $A \to F \to G$

step 3:

from 'G' we can reach only I

A* calculates.

$f(I) = g(I) + h(I)$

$= A \to F \to G \to I + h(I)$

$= (3 + 1 + 3) + 1$

$= 8$

$A \to F \to G \to I$ only one node I ..

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step 4 : I is having 3 possibilites H, E, J, G

A* Algorithm calculates.

$f(H) = g(H) + h(H) = (3+1+3+2) + 3 = 12$

$f(E) = g(E) + h(E) = (3+1+3+5) + 3 = 15$

$f(J) = g(J) + h(J) = (3+1+3+3) + 0 = 10$

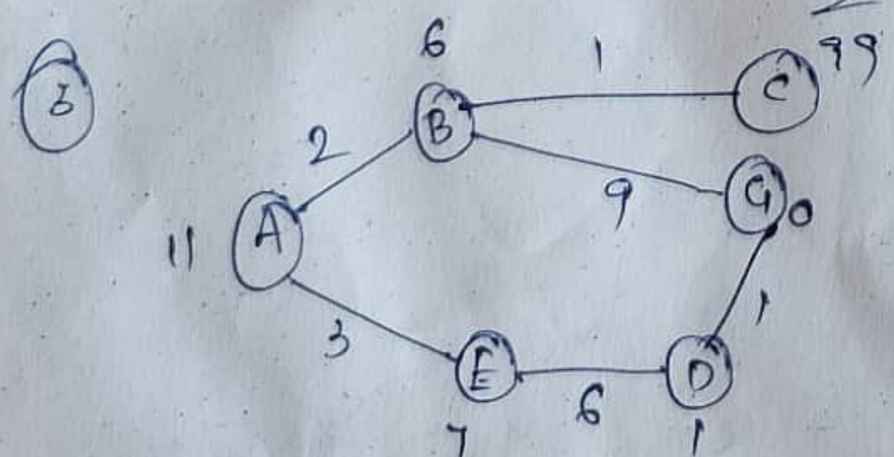
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f(J) is the minimum.

~~$f(G) = g(G) + h(G) =$~~

path A → F → G → I → J

2.



step 1 : A is the initial node

A is having two possibilities B and G

A → B A → G

A* algorithm calculates.

$f(B) = g(B) + h(B) = A \to B + h(B)$

$= 2 + 6 = 8$

$f(E) = g(E) + h(E) = A \to E + h(E)$

$= 3 + 7 = 10$

8 is minimum expand from B

step 2: B is having two successors C and G.

B → C and B → G

Can be represented as

A → B → C & A → B → G

f(C) = g(C) + h(C) = A → B → C + h(C)
= (2+1) + 99 = 102

f(G) = g(G) + h(G)
= A → B → G + h(G)
= (2+9) + 0 = 11

explore A → E → D → G

f(G) = g(G) + h(G)
= A → E → D → G + h(G)
= 3 + 6 + 1 + 0 = 10

So the final path is A → E → D → G

Heuristic functions:

Informed S.A (ISA)

These are a class of algorithms used in AI.

- It leverage additional information about the problem to guide the search procen more efficiently.
- ISA also known as heuristic SA or intelligent SA.
- Incarporate heuristic information
- utilize additional knowledge, represented by heuristic functions to make more informed decisions during the search procen, improving the efficiency of finding solutions in complex problem spaces.

Need for Heuristic functions:

In the vast landscapes of problem spaces, blind exploration is often impractical. This is where heuristic functions come into play.

They provide a way for AI syste
to make informed decisions aba
which paths to explore based on
heuristic evaluation.

Heuristic functions estimate the cost
from given state to the goal.
In informed search algorithms, thie
functions help prioritize paths
focusing on those likely to lead to
optimal solutions more efficiently.
They provide a measure of
optimism about a state and aid
in avoiding unnecessary exploration
of unpromising paths.
This characteristic ensures that the
search process is not only targeted
but also efficient.

calculate heuristic.

1. Define the problem
2. Identify state representations
3. Define the Goal
4. Identify Relevent metrics
5. Design the Heuristic function
6. Test and Refine.

4. Lack of Completeness.

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Year & Sem

Subject : AI

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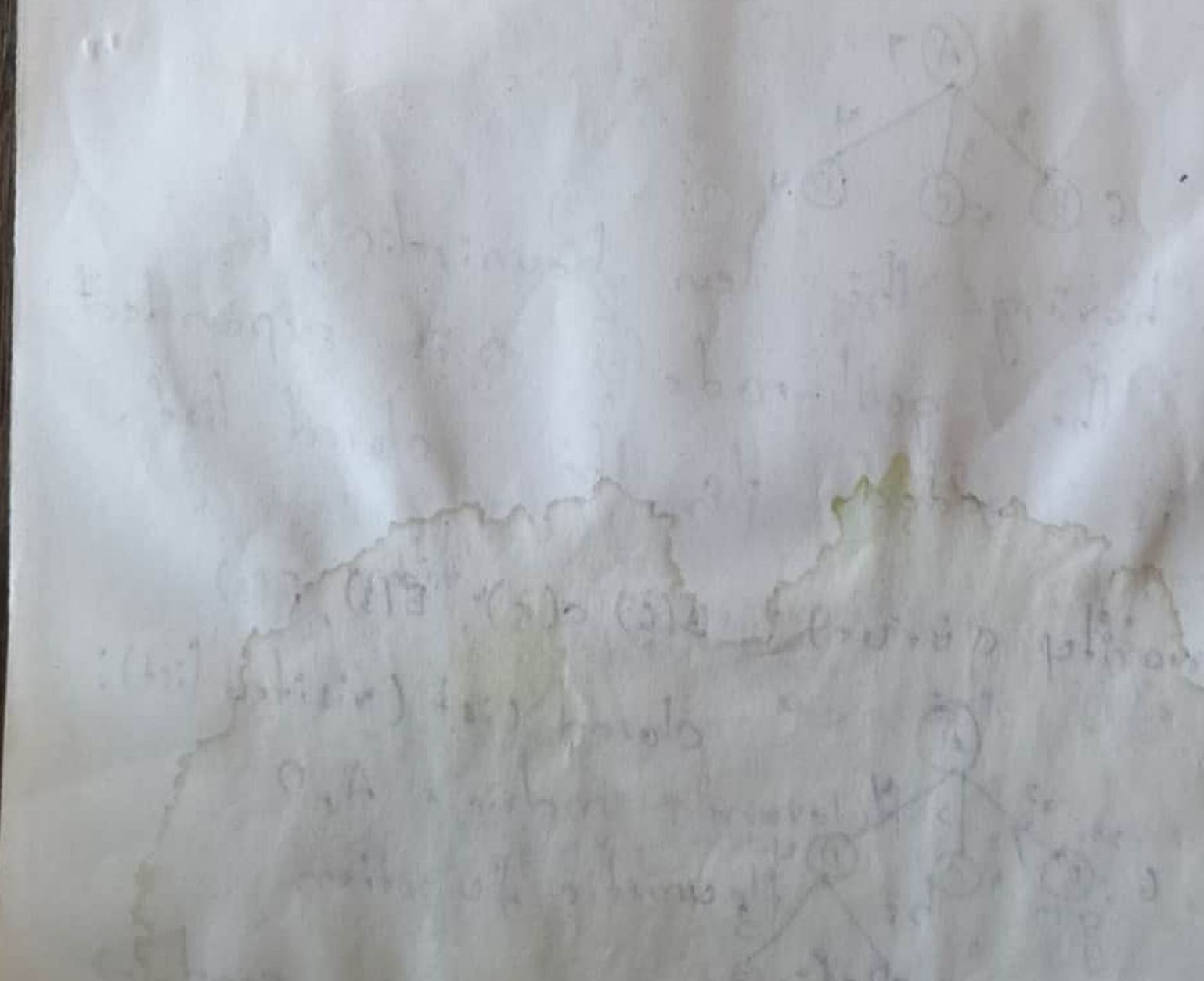
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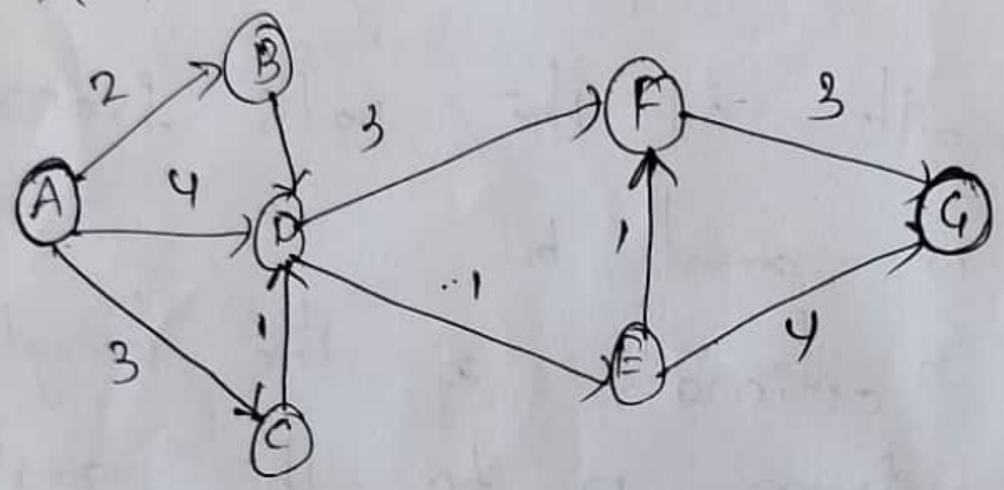
BFS is an instance of tree search and graph search algorithm in which node is selected for expansion based on the evaluation function $f(n) = g(n) + h(n)$

- $g(n)$ is the length of the path from the root to n and b
- $h(n)$ is an estimate of the length of the path from n to the goal node.

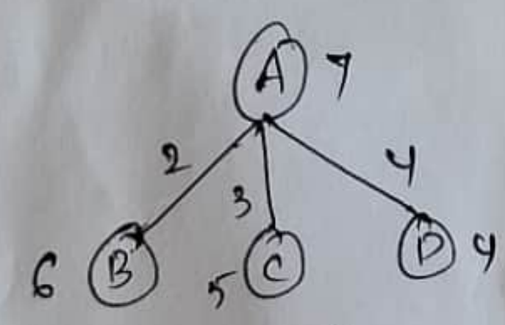
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- Consider the following directed graph having A as the starting node and G as the goal node. with edge cost as mentioned and the heuristic values for the nodes are given as
$h(A) = 7$, $h(B) = 6$, $h(C) = 5$, $h(D) = 4$,
$h(E) = 3$, $h(F) = 3$ $h(G) = 0$.

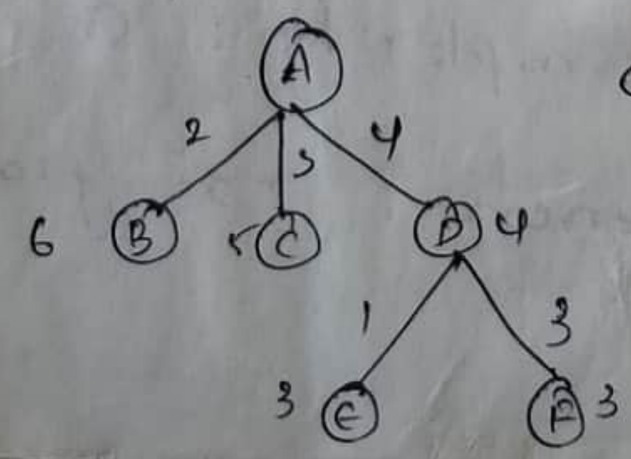


open list (priority queue) : B(6), D(4), C(5)
closed list (visited list) : A



node 'D' having the less heuristic,
D is not the goal node, D it expanded
and D is added to the closed list.

open list (priority queue) : B(6) C(5), E(3), F(3)
closed list (visited list):
A, D



cont →

cont → Example problem:

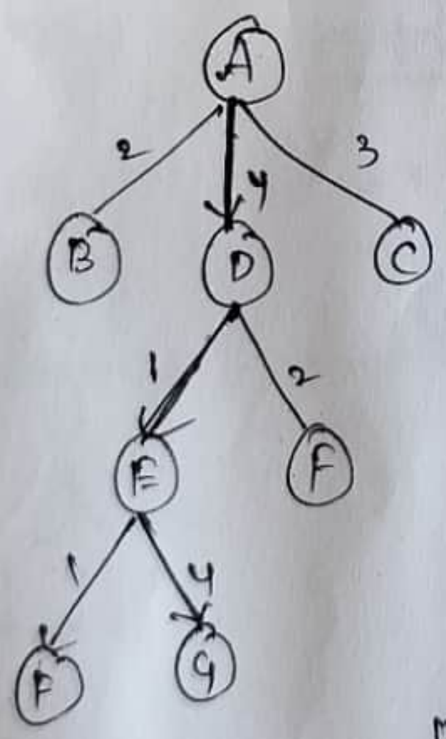
E and F are having same heuristic values, so we have to give preference based on alphabetic order.

E is given the priority.

open list (priority queue): B(6), C(5), ~~E(3)~~, F(3), G(0)

closed list (visited list): A, D, E, G.

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path: A → D → E → G.

path cost: 4 + 1 + 4 = 9.

GBFS uses only heuristic value to explore

- Time complexity: $O(b^m)$
- Space complexity: $O(b^m)$
- Completeness: not guaranteed.

OPEN : B T O E P

closed : C

---

key concepts:

. Heuristic function (h(n)

. Domain knowledge

. efficiency



Informed Search strategies in AI will use extra knowledge called ~~search strategies~~ heuristics

- Heuristic knowledge used to guide searches making them faster and more efficient than blind searches, by focusing on promising paths.
- Heuristic Searches Greedy BFS, A*, BFS all these depends on heuristic function h(n) to estimate goal (optimised).

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Key concepts for Informed Search strategies

. Heuristic function h(n) - H.f that estimates the cost from the current state to the goal state.

. Domain knowledge - uses problem-specific information (eg: map distances) to make informed decisions.

. Efficiency - reduces the search space compared to uninformed.

Informed search Algorithms will work by considering.

1. Evaluate neighbors - calculating heuristic values for all its neighbors.
2. prioritize: use heuristic (sometimes pathcost) to pick most promising neighbour node.
3. expand : Move to the chosen node and repeat the procen.

Advantages of Greedy BFS :

1. Simple and easy to implement
2. Fast & efficient
3. low memory requirements
4. flexible
5. effaiency - even in large search spaces is good

Disadvantages:

1. Inaccurate results — not always guaranteed to find the optimal solution if also concerned with finding most promising paths.
2. local optima - paths choosen is not best optimal paths
3. Heuristic function
4. lack of completeness.

Beyond classical search : involve moving part in systematic manner

local Search Algorithm : Instead of exploring the entire tree, these focus on current state & neighbours. local search algorithm, operate by searching from a start state to neighboring states.

- without keeping track of the paths nor the set of states that have been reached.

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- They are not systematic.
- They want explore a portion of search space, where a solution actually resides.
- They searches only the final state.
  - Hill climbing Algorithm
  - simulated Annealing
  - local Beam search
  - Evolutionary Algorithm - Genetic Algorithm.

optimal / feasible solution can be find using this.

physical Annealing

The simulated Annealing algorithm is based upon physical Annealing in real

- physical Annealing is the process of heating up a material until it reaches an annealing temperature.

- Simulated annealing (SA) is a probabilistic algorithm in AI that finds approximate solutions to optimization problems.

- SA is used when it is too expensive or impossible to find the best solution.

- SA works with a random solution
- Generate new solutions using a probability distribution.
- Accept or reject new solutions based on a probability.

- physical Annealing is the process of heating up a material until it reaches an annealing temperature
- It then cooled down slowly in order to change the material to a desired structure.

- when the material is hot, the molecular structure is weaker and is more susceptible to change.
- when the material cools down the molecular structure is harder and is less susceptible to change.
- Thermal dynamics equation calculates the probability that the energy magnitude will increase.

$$p(\Delta E) = e^{\frac{-\Delta E}{K * t}}$$

ΔE - energy magnitude

t - temperature

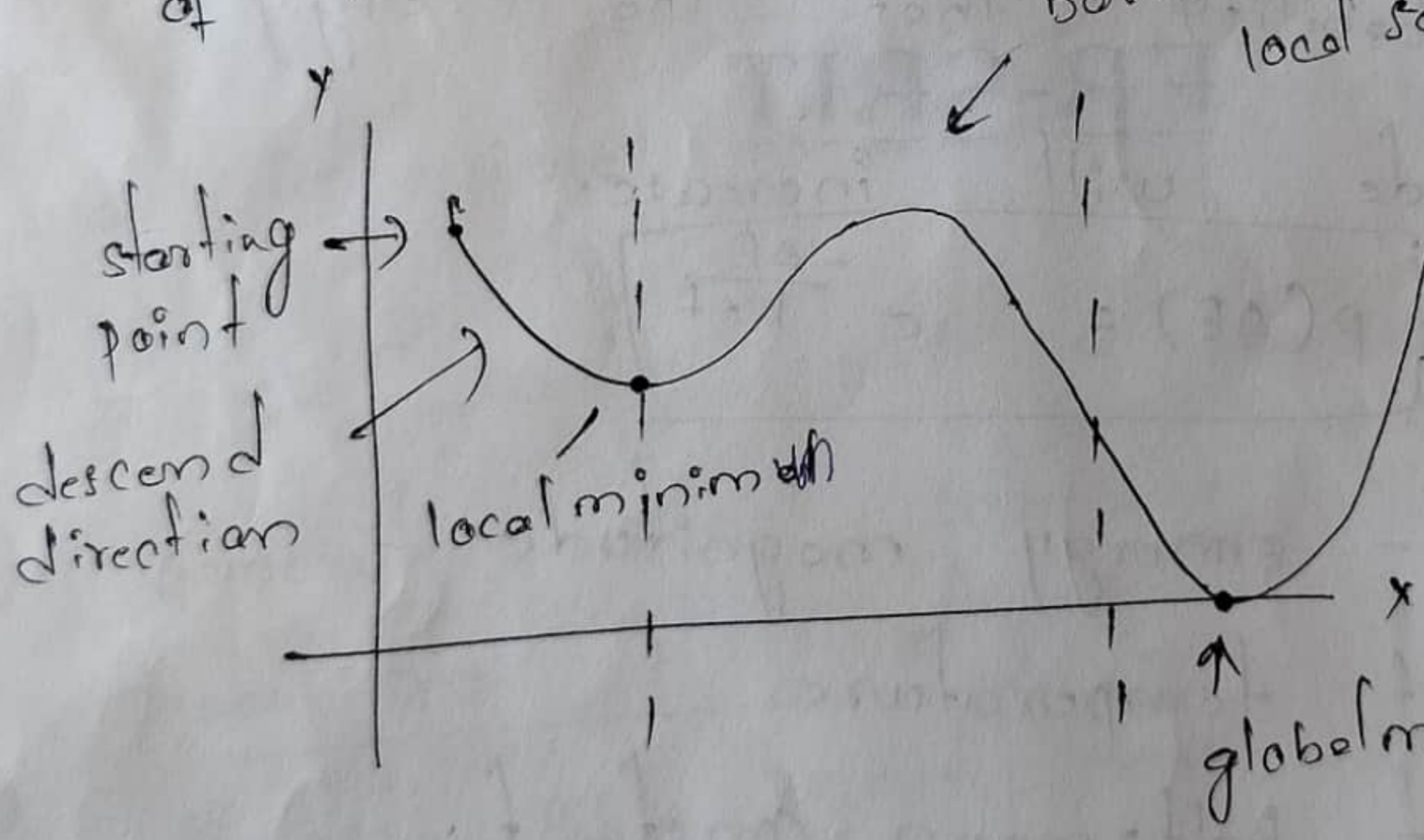
K - Boltzmann constant.

Simulated annealing is a stochastic global search optimization algorithm and it is modified version of stochastic hill climbing.

- It is suitable for nonlinear objective functions.

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- other local search do not operate well.
- The simulated annealing solution is to start by shaking hard (i.e at a high temperature).
- Then gradually reduce the intensity of the shaking (i.e lower the temperature)
- simulated Annealing is very useful for situations where there are a lot of local minima.



- Allows occasional uphill moves (non-optimal) to escape local optima.

- Beyond classical AI search which systematically explores states, techniques move to handle vast or unknown spaces using local search algorithms (Hillclimbing, simulated Annealing, Genetic AI)
- local search in AI for continuous spaces involves iterative optimization, moving from a current state solution to a neighbouring one in a vast solution landscape

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Hill climbing is a simple local search optimization algorithm in AI.
- It iteratively improves a single solution by moving to a better neighboring state until no further improvement is possible.
- Hill climbing works where each step is taken in the steepest upword direction possible until we reach a peak.

1. General Initial solution
2. Evaluate
3. move to better state
4. Repeat
5. Terminate

Types of Hill climbing

① simple Hill climbing
moves to the first neighbor that is better than the current state.

② steepest Ascent Hill climbing evaluates all neighbors and selects the one that offers the greatest improvement which is more thorough but computationally more expensive.

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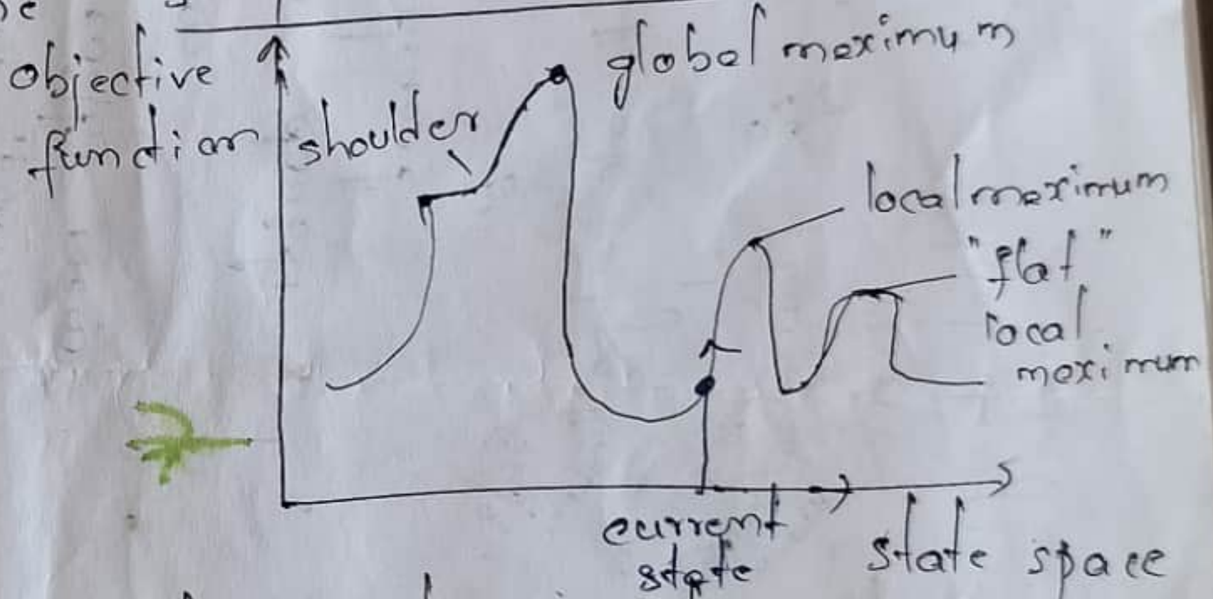
- stochastic hill climbing - Random selection of the step in improvement.
- Random Restart Hill climbing conducts multiple hill climbing searches from different randomly generated initial solutions to increase the chance of finding the global optimum.

Techniques used to overcome these issues include random restarts and combining hill climbing with other metaheuristic algorithms like simulated Annealing

- Hill-climbing Search Algorithm: 4/8/25. ②

It is a heuristic search algorithm which continuously moves in the direction of increasing value to find the peak of the mountain or best solution to the problem.

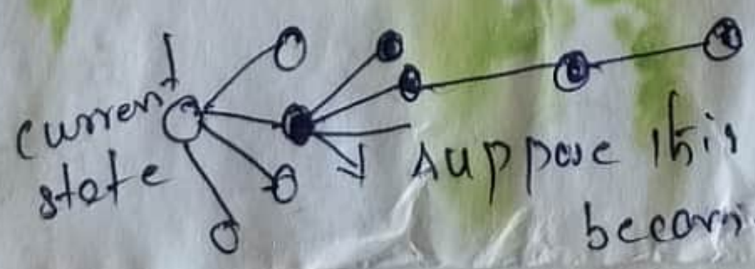
It keeps track of one current state and on each iteration moves to the neighboring state with highest value that is it heads in the direction that provides the steepest ascent.



- In this algorithm when it reaches a peak value where no neighbor has a higher value then it terminates.

It is also called greedy local search as it only searches its good immediate neighbor state and not beyond that.

Hill climbing is mostly used when a good heuristic is available.



suppose this is the higest in all its states it becomes an current state →)

→

- How many teams can I consider.
- After considering teams what work can I assign to them.
- How many members can I consider for a team.
- shall I go on with the same idea. if they come with new idea can I encourage them.
- If they are having any doubts can I ask them to interact with you directly.

Sabi
Cg. 2.20
~~Bweb~~ 10. BEE
OD. CSN
Wed, Thur
1 2:20

EEE - C
Fri 2.20
3:10.

Graphics
BwD.
In

(1) An IOT Based ~~system~~ Secured Vaccine Transportation System ~~for monitoring and tracking~~

(3) A COLLISION AVOIDANCE SYSTEM FOR VEHICLES.

1. Automated

content

4. Automation of

: 28 IK

Break Algorithm

flowchart process

Block diagram

T 3:10
F 3:10
2:20

1. Fogo
2.
3. App laundry (iron)

App
Faculty.

Different regions in the state space (states that considered in the system)
landspace (geographical area of land)

- local maximum: It is a state which is better than its neighbour states, but there is also another state which is higher than it.

Global maximum: is the best possible state of state space landscape. It has the highest value of objective function.

current state: is a state in a landscape diagram where an agent is currently present.

flat local maximum: is a flat space in the landspace where all the neighbor states of current states have the same values.

shoulder: is a plateau region which has an uphill edge.

- Hill-climbing search algorithm, which is the most basic local search technique. At each step the current node is replaced by the best neighbor.

function HILL-CLIMBING (problem)
returns a state that is a local max
current ← problem. INITIAL
while true do
neigbor ← a highest-valued successor
state of current
if VALUE(neigbor) ≤ VALUE(current)
then return current
current ← neighbor.

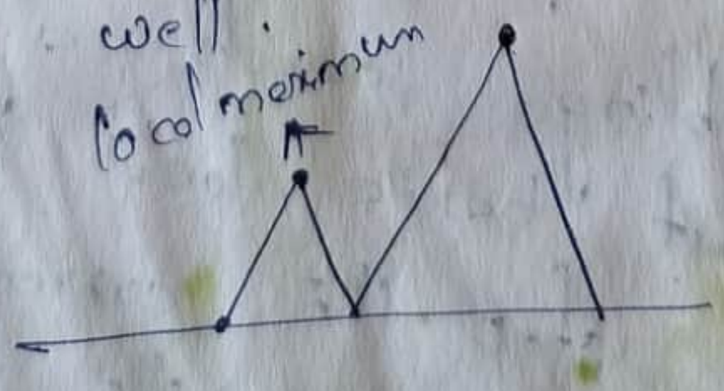
problems in Hill climbing Algorithm

1. local maximum:

A local meximum is a peak
state in the landscape which is
better than each of its neighboring states, but there is another state
also present higher than the local maximum.
which.

solution: Backtracking technique can
be a solution of the local maximum
in state space landscape.

create a list of the promising path
so that the algorithm can backtrack
the search space and explore other
paths as well.



Plateau : A plateau is the flat area of the search space in which all the neighbor states of the current state contains the same value.

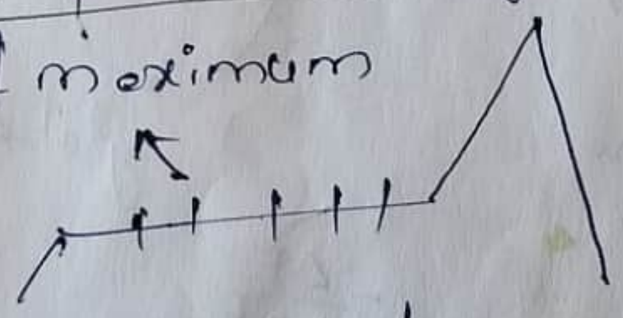
Because of this algorithm does not find any best direction to move.

- A hill climbing search might be lost in the plateau area.

solution : The solution for the plateau is to take big steps while searching, to solve the problem.

- Randomly select a state which is far away from the current state so if is possible the algorithm could not find non-plateau region.

plateau / flat maximum

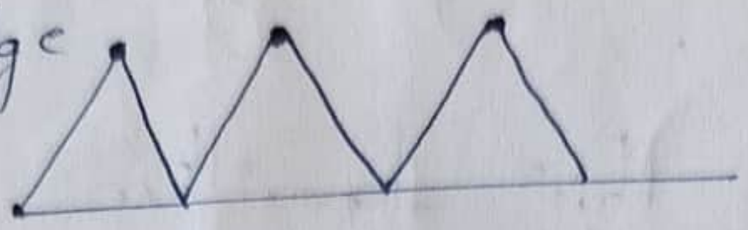


3. Ridges : A ridge is a special form of local maximum.

It has an area which is higher than its surrounding areas, but itself has a slope, and cannot be reached in a single move.

· solution: with the use of bidirec
search or by moving in different
directions we can improve this problem.

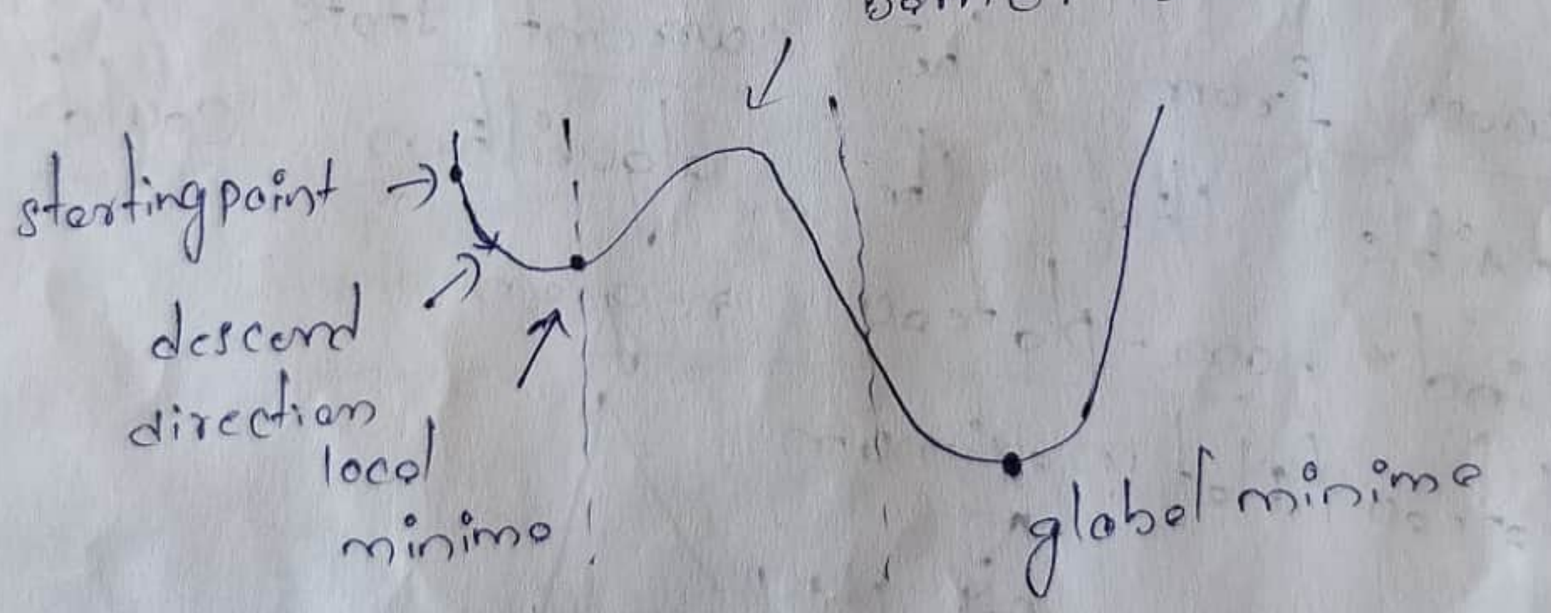
Ridge



- local heuristic focus on improving
solutions with in a limited neighborhood.
- Global heuristic aim to explore the
entire solution space for a better
potential, optimal solution.

physical Annealing

Simulated Annealing

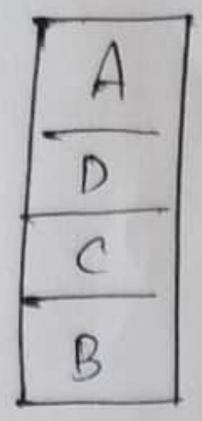


$$p(\Delta E) = e^{\frac{-\Delta E}{k * t}}$$

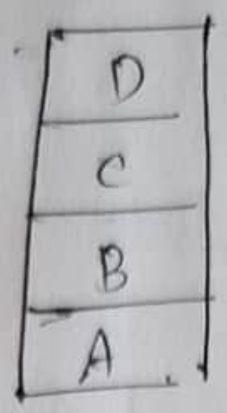
ΔE - energy magnitude

t - -

k - Boltzmann con



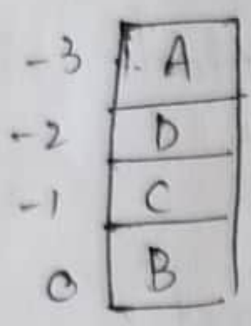
Initial state



Goal state.

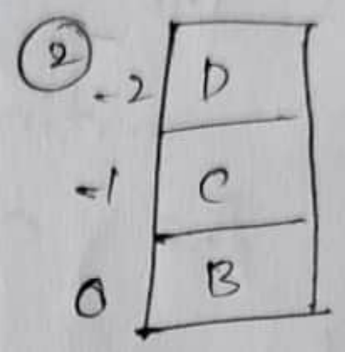
$h(1) = -6$

(1)



$h(2) = -3$

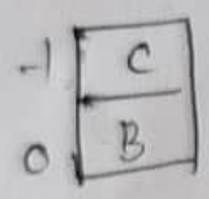
(2)



A .

(3)

$h(3) = -1$



A D .

(4)

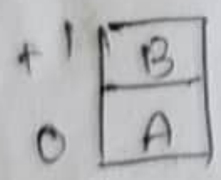
$h(4) = 0$

B A D C .

$h(6) = 3$

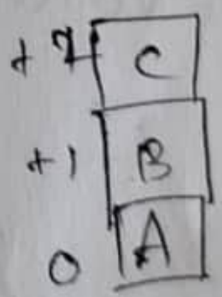
(5)

$h(5) = +1$



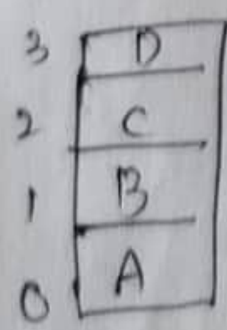
D C

(6)

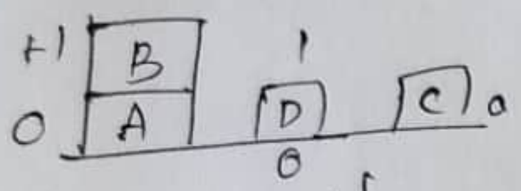


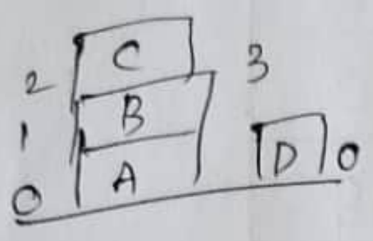
D .

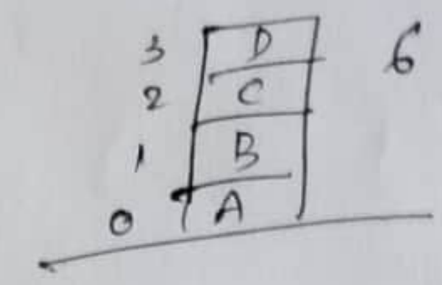
(7)



$h(7) = +6$.







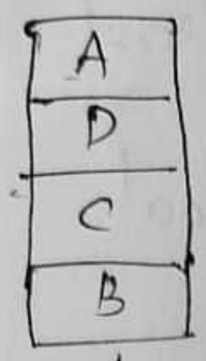
Algorithm

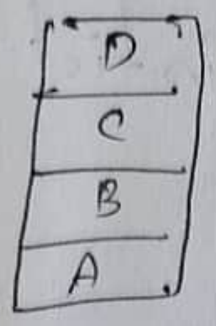
1. Evaluate the initial state.
   a. If it is the goal state then return it.
      otherwise continue with the initial state as the current state.
2. loop until the solution is found
   a. select an operator that has not yet been applied to the current state and apply it to produce new state.
   b. evaluate the new state
      i, If it is a goal state then return it and quit.
      ii, If it is not a goal state, if it is better than the current state, then make it as current state.
      iii, If it is not better than the current state, then continue in loop.

2.

climbing: local heuristic function

start





Goal

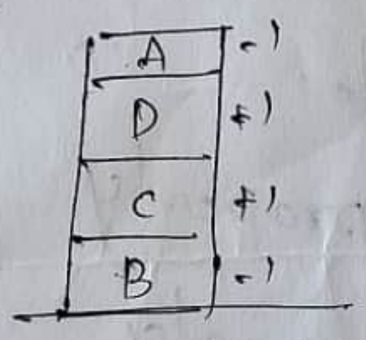
local heuristic:

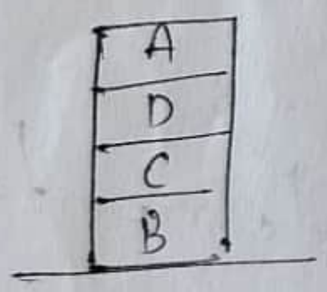
+1 for each block i.e resting on the thing if it supposed to be resting on.

-1 for each block i.e resting on a wrong block.

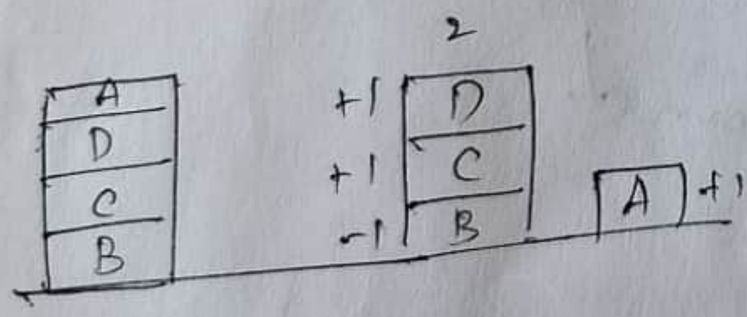
Blocks world

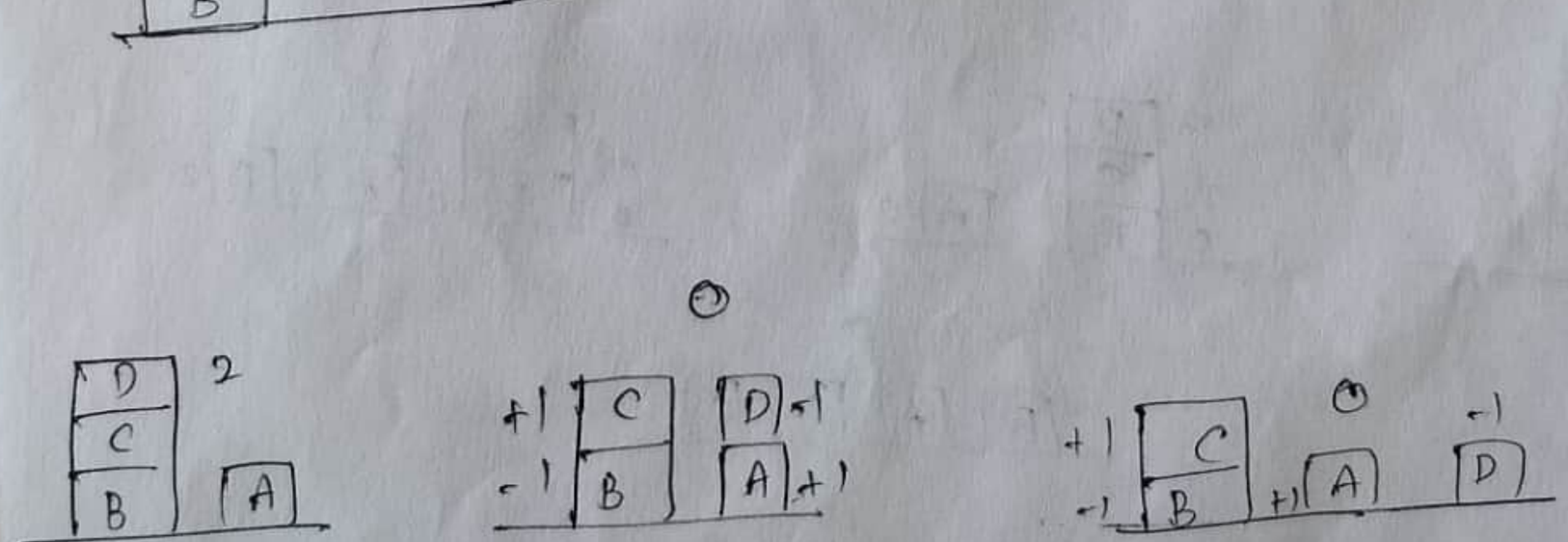
start





changing the state bringing A to down

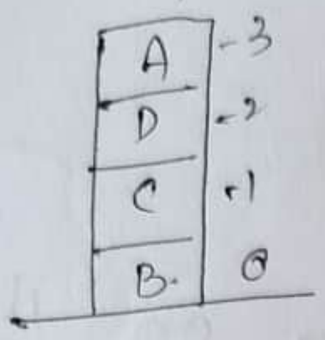


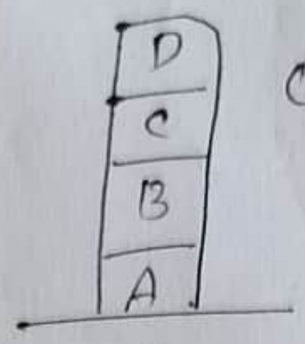


I am unable to reach the goal state.

So local heuristic is not optimal

- Global heuristic or Global function to reach from initial to final.





Blocks world.

Global heuristic:

for each block that has the correct support structure +1 to every block in the support structure.

for the given initial state below to final state.

B there are no nodes so 0

- C is not placed properly so -1 below one element
- D is not placed properly so -2 below 2 elements
- A is not placed correctly so -3 below 3 elements

- move A to Ground.

